EUROPEAN COMMISSION
DIRECTORATE-GENERAL FOR RESEARCH & INNOVATION

1.1. The project summary

Project Number [1]	720770	Project Acronym [2]	DAFIA

One form per project General information	
Project title [3]	Biomacromolecules from municipal solid bio-waste fractions and fish waste for high added value applications.
Starting date [4]	01/01/2017
Duration in months [5]	48
Call (part) identifier [6]	H2020-NMBP-BIO-2016
Topic	BIOTEC-02-2016 Bioconversion of non-agricultural waste into biomolecules for industrial applications
Fixed EC Keywords	
Free keywords	Municipal solids waste, Marine rest, fermentation, microbial, dicarboxylic acid, diamines, polyamides, nucleic acid, flame retardant, coating, gelatine, green chemistry.

Abstract [7]

Municipal solids waste (MSW) are collected by municipalities and represents more than 500 kg/capita (EU-27 average), 300 million tonnes overall every year in the EU-32. Currently, approximately 50% of this volume is landfilled. More than 1.3 million tonnes of Marine rest raw material (MRRM) are generated in Europe each year. Some countries, such as Norway and Denmark, have traditionally for animal feed. It will therefore be a challenge for the industry to develop methods to turn fish viscera and skin, currently considered as undesirable raw materials for hydrolysis and human consumption, into profitable products.
DAFIA will exploit MSW and MRRM as feedstocks for high value products. The parallel exploitation of the two feedstocks may create synergies. This expertise will be utilised in process development from MSW, while at the same time, new added-value products may be identified from both feed stocks.
The main objective of the DAFIA project is to explore the conversion routes of municipal solid waste (MSW), and marine rest raw-materials (MRRM) from the fish processing industries, to obtain high added value products, i.e. flame retardants, edible/barrier coatings and chemical building blocks (dicarboxylic acids and diamine) to produce polyamides and polyesters for a wide range industrial applications.
Different value-chains and products will be selected and explored based on the potential commercial value and the technical feasibility including new microbial strains and processes for conversion of major feedstock fractions, enzymatic and chemical modifications of components isolated from the feedstock or produced in microbial processes. Up to four cost-effective molecule groups suitable for the final selected applications will be targeted (nucleic acids, dicarboxylic acids, diamines and gelatine), & two value-chains (MSW & MRRM) will be evaluated at pilot scale to reach TRL5.

1.2. List of Beneficiaries

Project Number [1]	720770	Project Acronym [2]	DAFIA

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	AIMPLAS - ASOCIACION DE INVESTIGACION DE MATERIALES PLASTICOS Y CONEXAS	AIMPLAS	Spain	1	48
2	POLITECNICO DI TORINO	POLITO	Italy	1	48
3	SINTEF FISKERI OG HAVBRUK AS	SINTEF-FA	Norway	1	48
4	STIFTELSEN SINTEF	SINTEF-MC	Norway	1	48
5	DANMARKS TEKNISKE UNIVERSITET	DTU	Denmark	1	48
6	CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS	CNRS	France	1	48
7	NUTRIMAR AS	NUTRIMAR	Norway	1	48
8	INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL	IRIS	Spain	1	48
9	BIOTREND - INOVACAO E ENGENHARIA EM BIOTECNOLOGIA SA	BIOTREND	Portugal	1	48
10	DAREN LABORATORIES & SCIENTIFIC CONSULTANTS LTD	DLABS	Israel	1	48
11	MINE PLASTIK ENDUSTRIYEL URUNLER VE SERT MADEN KIMYA LIMITED SIRKETI	MINE	Turkey	1	48
12	BIO BASE EUROPE PILOT PLANT VZW	BBEPP	Belgium	1	48
13	BIOPOLIS SL	BIOPOLIS	Spain	1	48
14	ARKEMA FRANCE	ARKEMA	France	1	48
15	The National Non-Food Crops Centre	NNFCC	United Kingdom	1	48

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1. EXCELLENCE

1.1. OBJECTIVES

DAFIA will exploit Municipal solids waste (MSW) and Marine rest raw material (MRRM) as feedstock to produce high value products. The parallel exploitation of the two feedstocks will create synergies. Indeed, largescale industrial processes for separation of lipids and proteins and processes designed for isolation of other specific compounds have been implemented in the fish processing industry for several decades. As such downstream processing techniques are flexible and related expertise is adjustable, they will be utilised in process development for MSW, while at the same time, new added-value products may be identified from both feedstocks.

The main objective of the **DAFIA** project is to explore conversion routes of municipal solid waste (**MSW**), and marine rest raw-materials (**MRRM**) from the fish processing industries, to obtain **high added value products**, i.e. ***chemical building blocks*** (dicarboxylic acids and diamines) ***to produce polyamides for a wide range industrial applications***, **i.e.** ***flame retardants and edible/barrier coatings.***

Selected value-chains and products will be explored based on the potential commercial value and the technical feasibility. New microbial strains and cost-efficient processes for conversion of major feedstock fractions will be developed, as well as enzymatic and chemical modifications of components isolated from the feedstock or produced in microbial processes. Up to four molecule groups suitable for the selected final applications will be targeted (***nucleic acids, dicarboxylic acids, diamines, and gelatine***), and applications of co-products and residues will be proposed, to achieve a cascade valorisation of the residues. Two value-chains (***MSW & MRRM***) will be evaluated at pilot scale to reach TRL5 as shown in figure 1.1.a:

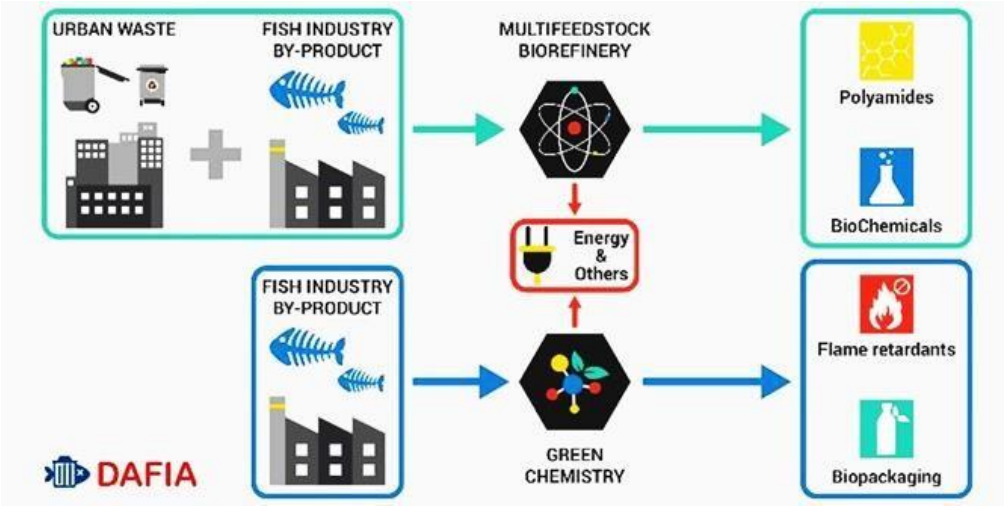


Figure 1.1.a. Scheme of DAFIA's project. The selected wastes for **DAFIA** project are:

1) Biogenic fraction of Municipal Solid Waste (MSW). MSW collected by municipalities, includes wastes from households, separately collected waste from commerce, offices and public institutions and other wastes of similar nature or composition, such as bulky food and garden waste. Its biogenic fraction (biodegradable garden and park waste, food and kitchen waste from households, restaurants, caterers and retail premises) ranges between 30-50% of total MSW, which represents more than 500 kg/capita (EU-27 average), 300 million tonnes overall every year in the EU-32[1], increasing at an annual rate of 10%. Currently, approximately 50% of this volume is landfilled, 30% is recycled (material separation and bio-waste recycling, including compost and digestion) and the remaining 20% is incinerated. In spite of the strong shift from MSW landfilling to recycling of materials such as plastic, wood or metals and energy production from the organic fraction (mainly by incineration), there is a challenge to turn the biogenic fraction into higher value products than bioenergy.

2) Marine rest raw material (MRRM). Biomass- Fishery Production EU-28 was more than 9.5 million tonnes live weight: hence, 2 million tonnes of MRRM are generated in Europe each year.[2] Some countries have traditionally utilised significant parts of the MRRM as silage, which is often processed into fish and animal feed or well, as a biofuel feedstock for anaerobic digesters. Only a small fraction of MRRM is used for human consumption or other value-added applications. In other countries, due to the lack of specialised infrastructure, MRRM are wasted or sent directly for animal feed without any attempt to extract the valuable components. For instance, cut-offs of fish species like herring and mackerel are wasted or sold for low value uses, such as the production of biogas, fuel or mink feed.

[1] European Environmental Agency report, 2013.
[2] Richardsen, et al. 2015. Analyse av tilgang og anvendelse for marint restråstoff i Norge, SINTEF Fisheries and Aquaculture, Trondheim.

Furthermore, thousands of tonnes of fish are discarded within European waters, due to low commercial value of the catches, lack of quota or because it has been required by law in the old common fisheries policy. New European directives introduce significant changes to these discarding practises.[3] The industry aims to increase the fractions of the MRRM that can be applied for human consumption. However, large amounts of by-products not suited for human applications will still be generated, and i*t will be a challenge for the industry to develop methods to convert these fractions into profitable products*. A Life Cycle Assessment (LCA) and a techno-economic viability assessment will also study the best approach to minimize the environmental impact of the new value chains without jeopardizing the cost effectiveness of the multi-feedstocks biorefinery operations.

Partners involved in the project have great experience and international reputations in all of the required materials and processing techniques, as well as world-leading market presence and interests. They have participated in many previous projects and industrial applications related to the use and production of chemicals from municipal and agroindustrial waste as shown in section 1.3.3. The specific **objectives and their means of verification** of the project are described in Table 1.1.a.

Table 1.1.a. Specific objectives of the **DAFIA** project, means of verification and partners involved[4].

	Objective and Means of verification (Main Partners involved)
Pre-treatment, extraction & purification of components from MSW & MRRM streams.	*To characterize the variability of the composition of biogenic fraction of MSW and develop strategies for preservation (BIOPOLIS).*
	To develop methods for pre-treatment and hydrolysis of MSW to be used as a fermentation substrate for production of chemical building blocks, and evaluate strategies for standardising the feedstock composition as a fermentation substrate (BIOPOLIS, DTU & SMC).
	To develop methods for extraction and purification of nucleic acids from MRRM (viscera) and from the microbial biomass generated as a by-product from fermentation, complying with a required purity of A260/A280 >1.6 and a yield of 12-15 g/kg MRRM (NUTRIMAR, SFA, SMC & POLITO).
	To ameliorate gelatine extraction technologies, improving textural characteristics in order to replace mammalian gelatines in food industry and employ the extracted products in the production of coatings. Target yields are calculated as 35 % gelatine from dry material in fish skin and backbone (NUTRIMAR & SFA).
Production of intermediate chemicals by fermentation.	*To design microbial hosts for production of muconic and adipic diacids and 1,5-pentanediamine from the carbohydrates of MSW, and protein and lipid fractions of MSW or RRM. Strains will be tested in industrial-like small-scale production system. Target production yields will be 5 g/kg MSW for the diacids, and >20 g/kg MSW for the diamine, with corresponding concentrations >2 g/l and >10 g/l, respectively (DTU, SMC, BIOPOLIS).*
	To develop robust fermentation processes, tolerating variations in feedstock composition, providing high volumetric productivities (>80 % of the productivities on glucose)- (DTU, SMC, BIOTREND, BBEU).
	To design novel and environmentally friendly processes for the recovery and purification of diacids and diamine from complex fermentation media at a competitive cost. Process developed at lab scale and implemented at pilot scale capable of producing high purity (>99% or less than 0.2% of monofunctional molecules) muconic acid, adipic acid and 1,5-pentanediamine (BIOTREND, BBEU).
	To develop fish gelatine-based coatings, optionally including MRRM based bioactive peptides, for application on fish fillet (edible coating) and fish packaging (barrier coating). Oxygen barrier improved by a factor up to 10 vs. base of current bioplastic packaging such as PLA or PHB, fish shelf life prolonged by a factor 2 (IRIS, NUTRIMAR & AIMPLAS).

[3] EU: Fisheries, Facts and figures on the Common Fisheries Policy 29/04/2015, http://ec.europa.eu/fisheries/reform/index_en.htm [4] Anexx II in section 4 summarizes the main expected conversion ratios of DAFIA products.

Production of end-products (coatings, additives and polymers).	*To modify the selected polymeric by-products for use as flame retardants, exploiting novel chemical routes. This includes improving the thermal resistance, fireproof, mechanical strength as well as other demanded properties. Mechanical, thermal and fire resistance tests will be performed. Loss of weight less than 2% at 300ºC by thermogravimetric analysis* (DLABS, ARKEMA, AIMPLAS).
	To design and optimize twin screw extruder parameters to produce thermoplastic compounds using standard polymers and bio-based polyamides and the DAFIA flame retardant. Bioplastics up to V1 classification UL94 on test specimens of 4 mm thickness will be designed (AIMPLAS & MINE).
	To produce esters, nitriles, alcohol, thiols or diols from muconic acid as monomers for polyamides. Biobased polyamides from adipic acid and 1,5-pentanediamine (polyamide 56) will be produced. Polyamides for coating applications with improved adhesion properties to metal and glass will be designed. Products will be obtained, having similar properties to commercial oil-based or sugar-based chemical groups (AIMPLAS & ARKEMA).

1.2. RELATION TO THE WORK PROGRAMME.

The DAFIA project is fully aligned with the challenge and scope addressed by the topic **BIOTEC-02-2016:** **"**Bioconversion of non-agricultural waste into biomolecules for industrial applications" as shown in Table 1.2.a.

Table 1.2.a. Relation of the **DAFIA** project solutions to challenge and scope described at topic BIOTEC02:2016.

Challenges & Scope addressed by topic.	**Solutions proposed by DAFIA project**
To develop biotechnology approaches for the conversion of municipal solid bio-waste fractions and sludgy biowaste from other industries, into higher value bio-based products, such as chemicals and chemical building blocks, biopolymers, materials and bioactive compounds. Physico-chemical technologies concomitant to the enzymatic/microbial processes are also needed.	Two huge waste were selected; i.e. MSW and MRRM to obtain the products detailed in table 1.1. Biotechnological approaches are essential to achieve the goals, involving enzymatic hydrolysis of the feedstocks, development of microbial strains designed for conversion of the feedstocks to diamines and diacids, and development efficient fermentation processes. Intermediates will be chemically modified to obtain the final product. The applications for MSW are targeted to bulk lower cost products while the applications for MRRM target lower volume/higher value products.
This includes sustainable downstream steps for product separation and purification.	Innovative, environmentally benign extraction and purification processes will be used for isolation of the targeted products from biomass and the fermentation broths. When possible, enzymatic processes will be preferred to chemical ones and opportunities for integration with the subsequent processing steps will be exploited. Solvents and other chemicals will be recycled.
Address the current technical state-ofthe-art regarding waste utilisation for bio-based-products taking into account the current market and legislative barriers in the EU.	Detailed reviews of current EU policy and member state legislation coupled with stakeholder engagement and interview will enable the identification of non-technical barriers to product and market development. Furthermore, DAFIA aims to go one step further and investigate the potential for future legislation and standards, develop several scenarios and propose a road map for future legislation. Task 8.5 will monitor the current and future market and legislative barriers during the project life mainly the products used for direct human use.

The feasibility of integrating the newly developed approach into existing value chains should be assessed and demonstrated. Proposals should have a strong industry drive and prove the techno-economic viability of the proposed value chain.	10 of 16 are industrial partners, covering the full value chain of the selected products (see figure 2.1 b in impact section). The current low commercial value of the selected waste, the efficient and affordable selected downstream processing, and the high value of the target products allow justifying the economic viability of the proposed value chains. Based on current information (TRL 3), an initial *technoeconomic viability has been detailed in section 2. In WP8 and WP9, the techno-economic viability of the develop products and technologies will be refined as well as the plans for exploitation and business plan*
They need to also take into account the optimisation of the final product's "end of life" through, for example, biodegradation or recycling.	Proposed products will be fully recyclable. The flame retardants, diacids and diamines/polyamides will have the same behaviour as current natural or petrol based analogues. The obtained barrier coatings will provide recyclable and/or compostable packaging solutions.
A life cycle assessment of the entire value chain should be included.	A full LCA will be proposed in WP8. DTU and AIMPLAS have broad expertise in LCA studies.

1.3. CONCEPT AND METHODOLOGY

1.3.1. Overall concept

Implementation of environmental policies, especially waste policies, is one of the European Commission's key priorities, as confirmed by its proposal for a 7th Environment Action Programme (EC, 2012) and the Roadmap to a resource efficient Europe (EC, 2011). While the EU's Waste Framework Directive (EU, 2008) and Landfill Directive (EU, 1999) set binding targets for recycling organic waste and diverting biodegradable municipal waste from landfill. With the world population set to approach an estimated 9 billion by 2050, against a background of finite natural resources, Europe needs renewable biological resources - not just for securing healthy food and animal feedstuffs but also for raw materials to produce industrial products and bio-fuels. DAFIA is in accordance with these priorities by designing a cascade approach in order to maximize the value of each components of the MRRM and MSW.

On 13 February 2012, the European Commission adopted a strategy for «Innovating for Sustainable Growth: A Bioeconomy for Europe»[4]. A strong bioeconomy will help Europe to live within its limits. The sustainable production and exploitation of biological resources will allow the production of more from less, including from waste, while limiting negative impacts on the environment and reducing the heavy dependency on fossil resources, mitigating climate change and moving Europe towards a post-petroleum society

MSW biowaste contains large amounts of polysaccharides (30-40% cellulose, hemicellulose, starch and pectin), lipids (10-15%) and protein (5-15%), depending on the origin and pre-processing steps. Efficient conversions of mixed, and variable feedstocks are challenging. To date, mixed wastes have been transformed by thermochemical processes, such as hydrothermal liquefaction (HTL) and gasification, and by anaerobic digestion, leading to methane and bio-hydrogen, fuels (ethanol and butanol), compost and fertilizers. Production of more complex chemicals is limited by large-scale economic feasibility due to pre-treatment and conditioning costs of the recalcitrant raw materials. One approach to overcome this challenge is to use the gases obtained by thermochemical processing and/or anaerobic digestion, as substrates for fermentation to other products, as section 1.4.1 details. DAFIA's approach, which will be achievable in a shorter term, is to develop more cost-efficient pre-treatment and fractionation methods that will open for the use of MSW as a general fermentation substrate, utilising either the carbohydrates only, or the complete organic fraction.

DAFIA aims to establish new technologies for pre-processing and fractionation of MSW for microbial conversions, to develop microbial strains able to convert MSW to the targeted intermediate products with high productivities, and for their subsequent transformation into marketable products.

Direct extraction of protein and lipids for conversion to other, added-value products to further improve the process economy, will also be explored. For instance, lipids can be used as biofuels or biolubricants; hydrolyzed proteins can be used as biofertilizers; biosurfactants can be obtained from both lipids and aminoacids.

Fish rest raw materials consist of head, tails, skin, entrails, fins and frames, containing mainly proteins, fats and minerals. The composition varies with fish species (fat or lean fish, e.g. salmon and mackerel *vs.* cod) and fraction. For instance, mixed wastes from a Spanish fishmonger (mainly lean, white-fish), contained 58 % proteins, 19 % lipids, and minerals[5], while mixed salmon waste contain 45-50 % lipids and 35-40 % protein[6]. Fish filleting and

[4] http://ec.europa.eu/research/bioeconomy/index.cfm?pg=policy&lib=bbpp
[5] Ghaly et al., Fish processing wastes as a potential source of proteins, amino acids and oils, J Microb Biochem Technol 2013, 5:4
[6] Ranirez (2007) Salmon by-product proteins. FAO Fisheries Circular No. 1027

processing companies receive food grade fish, and if the fish is treated hygienically, trimmings, heads and backbones, which together accounts for up to 50 % of the fish[7], and 60-70 % of the RRM, are a source of valuable marine oils and proteins that can be applied for food, nutraceuticals and cosmetics[8]. The industry aims at increasing the fractions of by-products that can be applied for the high-value, human markets. Skin, which currently is underutilised, and also heads and bones, are rich in collagen and a good source for gelatine extraction. Nevertheless, large amounts of byproducts not suited for human applications are generated. Viscera, with guts, liver and roe as the major fractions, constitute 30-40 % of the RRM. The guts, and unsorted entrails containing guts, are undesired as raw materials for human consumption, mostly due to ethical considerations. Viscera are also a perishable raw material, difficult to handle. However, it is rich in protein and active endogenous enzymes, which could be employed for protein hydrolysis and solubilisation. In addition, viscera contain roe and milt, which are excellent raw materials for extraction of nucleic acids.

DAFIA aims to establish new technologies for extraction of collagen and nucleic acids from fish RRM, and further chemical conversion of these to the targeted, higher value products.

DAFIA will employ active, endogenous enzymes from viscera for more sustainable hydrolysis of protein, to facilitate fractionation and extraction of nucleic acids, and to generate bioactive peptides from the protein residues. Depending on the origin, these can be applied in the edible gelatine coatings, in animal feed, or as a nitrogen and nutrient source in the fermentation media. We will seek for improvement of the characteristics of fish gelatine, to substitute mammalian gelatines.

Although DAFIA will not allow valorising each single compound in the target waste, its cascade approach focuses on those of greatest potential. In addition, the possible use of remaining fractions will be suggested (based on LCA and techno-economic analyses results) for post project further improvements of the overall process economic and environmental sustainability, non-useful components could be employed to produce biogas or other type of energy. Furthermore, a stakeholder group and networking with other EU projects will be proposed to select the best technologies to valorise the separated components from MSW and MRRM.

Figure 1.3.a shows the target products, processed involved and their selected feedstocks. Higher-value products (nucleic acid, peptine, gelatine, etc..) will be the prioritized application of lipids and proteins from fish by-products. The others components can be used as fermentation substrate. Detailed information is presented in section 1.4.

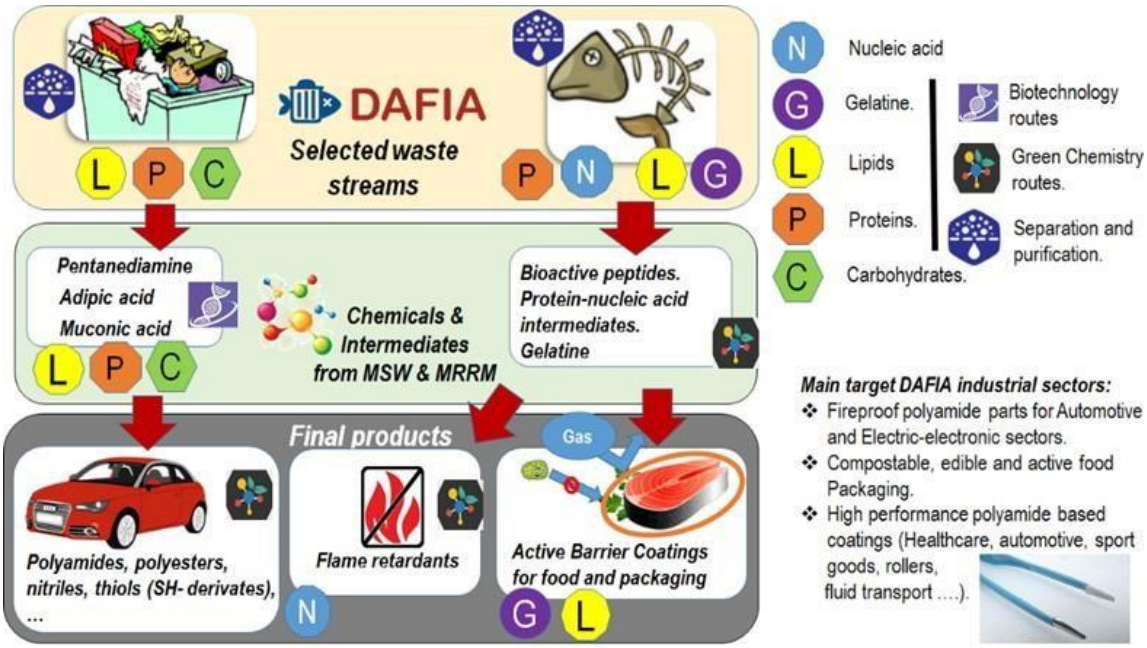


Figure 1.3.a DAFIA Flow Diagram including partners and sectors involved in each step.

1.3.2. Positioning of the Project.

The activities developed in **DAFIA** project will be implemented at TRL3 with a TRL target of 5. Most of the proposed innovations are based on technologies validated at laboratory level, as it is shown in Table 1.3.a:

Table 1.3.a. TRL justification of **DAFIA** developments.

[7] Guerard F, Sellos D, Le Ga Y. Fish and shellfish upgrading, traceability. Adv Biochem Eng Biot 2005;96:127-63.

[8] Torres et al, By-products from seafood processing streams. In: Maximising the Value of Marine By-Products (edited by F. Shahidi), 2007.

Technology proposed	Current→ target TRL	Justify current TRL
Extraction of nucleic acids from protein rich biomass	3→5	POLITO has already set a lab-scale method for the recovery of nucleic acids at yields (8-10 %) and their purity is comparable with that of commercially available products from chemical suppliers.
Microbial conversion	3 →4/5	DTU, BIOPOLIS and SMC have a strong background in designing of microbial host strains for chemical building blocks production, and several diacids and diamines producing strains are available or have been reported in the literature. Typically, the production is mostly done from commercial C6 sugars, and the production of the target diacids and diamines from multicomponent biogenic municipal waste is a novel approach. The microbial strains obtained in the lab will be transferred to process development & scale-up activities.
Bioprocess design for production of diacids and diamines from complex sources,	3 → 4/5	BIOTREND and SMC have up-to-date infrastructure for the development and optimization of bioprocesses, and wide experience in developing processes involving complex raw materials. Specific work will be carried out to be able to cope with the raw material(s) of interest and their fluctuation without detrimentally affecting the process robustness.
Fermentation scale-up	3 → 5	BBEU operates a multi-purpose pilot plant allowing adapted process scale-up which mimics an industrial setup. Its fermentation capacity ranges from 10 L to 15.000 L. The fermentation scale of the most promising product will be 1000 L (1500 L fermenter).
Downstream processing	3 → 5	BIOTREND and BBEU have been developing processes for the recovery of various biomolecules, including diacids (BIOTREND in EU project BIOREFINE-2G) and molecules similar to diamines such as amino acids. Further BBEU operates a multipurpose pilot plant, which will be used to scale-up and demonstrate the functionality of the developed downstream processes. Possible issues on malodour caused by diamine are taken into account in the setup of the fermentation hall, and can be further reduced by installing additional activated carbon filters.
Diacids conversion to polymers	3 →5	AIMPLAS is developing polymers from several diacids and diamines from other sources in projects such as EU project BIOREFINE-2G and national and private
		projects. Arkema is a producer of Sebacic acid (biobased C10 diacid) and uses it to produce decanediamine and in its polyamides[9].
Green Flame retardants	3→5	In BRIGIT EU project, DLABs has developed a flame retardant based on chemical modifications of lignosulfonates, a complex polymer, DAFIA will use similar chemistry.
Edible & barrier coatings for food and packaging.	3→5	Coating formulations based on proteins (whey, gelatine and others) have been validated at lab scale at IRIS and in several previous EU projects (OliPHA and Wheylayer).
Bioconversion of MSW and agrofood derived by-products	3→5	BIOPOLIS has developed and validated physico-chemical and biochemical processes for the optimal conversion of different fractions of MSW and agro-food derived by-products into valuable organic compounds.

1.3.3. Research and innovation activities linked with DAFIA.

We were unable to identify public European projects with the same research objectives as those proposed in **DAFIA**. However, there are a number of national and European projects with strong synergy with **DAFIA**, and details are summarized in Table 1.3.b.

Table 1.3.b. Partners' projects related to **DAFIA**.

Description of experience/project	Scope	End Year	Website for information (DAFIA's Partner)
BUGWORKERS: New tailor-made PHB-based nanocomposites for high performance applications.	FP7-NMP	2014	www.bugworkersproject.eu/ (AIMPLAS, BIOTREND).

[9] www.arkema.com/en/products/solutions/biobased/; www.arkema.com/en/products/product-finder/range-viewer/Hiprolon/

BRIGIT: New biopolymers produced from lignocellulosic residues for flame retardancy highly demanding applications.	FP7-KBBE	2016	www.brigit-project.eu (DLAB,AIMPLAS,BIOTREND).
BIOREFINE-2G: Production of Dicarboxylic Acids and Bio-based Polymers**.**	FP7-KBBE	2017	www.biorefine2g.eu/ (DTU, AIMPLAS, BIOTREND).
OLI-PHA A novel and efficient method for the production of PHA polymer-based packaging from olive oil waste water	FP7-NMP	2015	www.olipha.eu (IRIS)
WHEYLAYER2: Barrier polymers for sustainable packaging	FP7-SME	2014	www.wheylayer.eu (IRIS)
APROPOS: Added value from high oil and high protein industrial co-streams – collaborative research project	FP 7 KBBE	2014	www.euapropos.eu (SFA)
TRANSBIO: Innovative cascading concept for the betterment of agro-industrial sub-products to obtain biopolymers.	FP7 KBBE	2015	www.transbio.eu (BIOTREND)
FUNGUSCHAIN: Valorization of mushroom agro-wastes to obtain high value products	H2020/ BBI	2020	(BIOTREND)
EuroBioRef: European Multilevel Integrated Biorefinery Design for Sustainable Biomass Processing.	FP7	2014	www.eurobioref.org (ARKEMA)
ThermoFactories: Thermophilic cell factories for efficient conversion of brown algae biomass to high-value chemicals	ERA-NET	2019	www.marinebiotech.eu/ (BIOTREND, SMC & DTU).
3R2020+: From waste to resource by recycling	Spain level project	2019	//3r2020.com/ (BIOPOLIS)
PHARSU: Design of new process for conversion of organic substrates from urban waste flows to bioplastics		2013	(BIOPOLIS)

A thorough freedom to operate search has been done on existing patents and technical articles. EU and world-wide data bases have been consulted and there are no patents invalidating the proposed DAFIA project objectives. Besides product innovations, the innovation potential for the partners regarding the improvement of the municipal solid waste and fish waste fractionation and purification, microbiotechnological processes, downstream, catalysis, green chemical and final production processes is substantial.

1.3.4. Approach and methodology.

DAFIA is a **4 year project**, structured into 10 work packages. **The DAFIA** project will take into account all the development steps to obtain valuable products from low value by-products to fit real industrial needs, including a work plan with a well- balanced structure that will combine research activities for selection, extraction and purification of intermediates (WP1 and WP2), and production of valuable products (WP3 to WP6) with piloting activities inWP7 and environmental and economic studies in WP8. DAFIA is a well-balanced consortium covering all the value chain from waste valorisation to end-users. In WP1 and WP2, the involved partners will lead the separation of the valuable components of MSW and MRRM based on the requirements defined by WP3 to WP5 partners. Applications will be proposed for the non-valorised components of both waste sources.

DTU will lead the production of diacids from MSW and MRRM, with SMC will be responsible for the diamine production. SMC and BIOTREND will contribute in the selection of the strains and to optimize the fermentation processing. BIOTREND and BBEU will study the best available technologies for downstream processing. Diacids and diamines need a high purity, but they must be cost-competitive in comparison with current bio-based products coming from sugars, vegetable proteins or oils, but also having in mind that widespread market penetration will only occur only if they can also compete with non-biobased products currently used by the chemical industry.

In parallel, in WP4, DLABs and POLITO will produce thermal resistance nucleic acid based flame retardants (from some components of MRRM). For by-product fish gelatine, IRIS (WP5) will prepare an edible and/or barrier coatings for fish and packaging for fish products and other foods that require barrier properties. In WP6, AIMPLAS will be in charge to polymerize PA56 (copolymer of 1,5-pentanediamine and adipic acid) using the monomers produced in WP3 by DTU and collaborators. ARKEMA and CNRS will study the production of several high added value products coming from muconic acid including thiol and nitrile derivatives. Finally in this WPs, several flame retardant compounds will be developed by combining the flame retardant additives produced by DLABs in WP4 and the different bio-based polyamides produced in WP6. The new compounds will be validated mainly for the automotive sector using the most accepted fireproof standards used by the main car manufacturers. All the products described in previous WPs will be piloted in WP7, mainly by the industrial partners.

The environmental, LCA analyses, safety, standardization and economic assessment to validate the main objectives of the **DAFIA** project will be carried out in WP8 leaded by BIOTREND with the support of all partners. Innovation management, business plan and Communication, Dissemination & Exploitation activities regarding the project results will be done in WP9 leaded by MINE. NNFCC partner will be in charge of market analysis and elaborate the preliminary business plan for all the DAFIA partners and initial contacts (Stakeholders) with companies outside of DAFIA consortium interested in the project results. WP10 will be dedicated to management and risk assessment.

1.3.5. Sex and Gender analysis.

The technologies, processes and materials developed in DAFIA are gender-independent. All DAFIA Partners have internal gender equality jobs creation and promotion policies. The consortium will foster the presence of women in the technical and management teams and decision board implemented to govern DAFIA project. The coordinator of DAFIA is Ms. Carracedo. Project aims to contribute to these objectives for enhancing Gender Equality by involving numerous female researchers (see section 4) in the project and by and by promoting women researchers to play a leading role in work packages. Actually, WP1 (Dr. Marta Tortajada), WP2 (Dr. Rasa Slizyte), WP3 (Dr. Irina Borodina), WP5 (Dr. Elodie Bourgnicourt), WP9 (Ms. Nalan Onar) and WP10 (Ms. Itziar Carracedo) are managed by female researchers, and as it is described in section 4, near to 50% of the staff researchers involved in the project are women.

The Treaty on European Union, the European policy establishes equal opportunities between women and men. In particular articles 2 and 3 establish equality between women and men as a specific task of the Community and also to promote equality. Monitoring the progress toward gender equality in science has become a well-established activity of the European Union research policy. In accordance with these principles, the approach of DAFIA is fully in line with Directive 2002/73/EC of the European Parliament and of the Council of 23 September 2002 on the implementation of the principle of equal treatment for men and women as regards access to employment, vocational training and promotion, and working conditions.

1.4. AMBITION.

1.4.1 State-of-the-Art and progress in handling, processing and fractionation of organic fraction of Municipal solid waste (MSW).

Technology developments for treatments of the organic fraction of municipal solid waste have so far been aimed at improving degradability and conversion to biogas and compost as the main outputs of non-recycled bio-waste[10-11]. This kind of operations include chemical (ozonisation, peroxidation, alkali and acid addition), physical (mechanical, thermal, high-pressure, microwave irradiation, ultrasound, aeration) and also biological pre-treatments to enhance bioavailability and solubilisation of target compounds and reduce undesired processes, such as uncontrolled fermentation by contaminating microorganisms[12], or degradation.

Furthermore, there are several projects aiming to make biofuels from MSW and animal wastes. Some of them would feed either the waste to microorganisms, or would first go through a hydrothermal liquefaction process or a gasification, followed by fermentation of gases to the desired products. Examples of previous projects include: PERSEO Bioethanol® process, which has developed a fermentative process for the transformation of the organic fraction of MSW to bioethanol and bioproducts. Lanzatech and Ineos Bio have projects to gasify biomass/municipal wastes to make ethanol (and then other products). Enerkem has full scale and pilot plants to gasify municipal wastes and produce methanol and ethanol and they are building a network in the Netherlands (Akzo-Nobel) for technology alliances. Renmatix is developing supercritical water treatment of lignocellulose to extract sugars for fermentation. *Synergy with current or past fuel -related projects will be assessed in DAFIA project*.

Moreover, the purpose of pre-treatment is to facilitate extraction of different fractions required for further revalorization, and to prepare MSW biomass for optimal fermentation. The pre-treatment will be essential to obtain a cost-efficient fermentation and downstream process. So far the purpose of pre-treatment has been to overcome the intrinsic characteristics of biomass, including recalcitrance, heterogeneity, multicomposition, and diversity.[13] Though some of these approaches have proven to be valuable to increase the availability of polysaccharides to enzymatic hydrolysis and fermentation to biofuels, a specific and optimized process must be developed. Additional effort will be put on the lipid and protein fraction, as biomass pre-treatment procedures have mostly been developed for cellulose or lignin components. There is no an ideal pre-treatment procedure, therefore, attending to the requirements of the different fractions for further revalorization, combination of different pre-treatments will be applied. The chemical,

[10] www.epa.ie/pubs/advice/waste/municipalwaste/EPA_MSW_Pre-Treatment_Guide_final%20Amended.pdf

[11] Montejo C., et al. Influence of input material and operational performance on the physical and chemical properties of MSW compost, J. Environmental Management 162 (2015).

[12] Solubilised components that can be used as carbon source could have undesired microorganisms growing and reducing the content of the carbon source for the intended process.

[13] Cheng H.Z., Liu Z-H. Steam explosion and its combinatorial pretreatment refining technology of plant biomass to bio-based products. Biotechnol. J., 2015, 10 (2015), 866-885.

physico-chemical[14] and biological[15] pre-treatment processes to be employed in DAFIA are summarized in table 1.4.a.

Table 1.4.a. Major advantages and disadvantages of different pretreatment processes[16]

Pre-treatment methods	Major advantages	Major disadvantages
Acid	Reduces hemicelluloses content. Low cost.	Chemical recovery limitations.
Alkali	Reduces content of hemicelluloses and lignin. Low cost.	High pollution and high chemical recovery cost.
AFEX or ARP[18]	Causes cellulose swell and increases accessible specific surface area.	High equipment and ammonia cost.
Hydrothermal	Solubilizes hemicelluloses and recovers sugars from hemicelluloses.	High equipment cost.
Steam explosion	Solubilizes hemicelluloses, alters the structure of lignin, cost effective.	High equipment cost and generation of inhibitors.
Ion liquid	Reduces crystallinity.	High cost of Ion Liquids.
Oxidative	Efficient removal of lignin.	High cost of reactive.
Organosolv	Separation and recovery of high quality lignin.	High price of organic solvents.
Milling	Increases accessible specific surface and reduces crystallinity.	High energy consumption.
Biological Delignification	Degrades lignin and hemicelluloses.	Long pre-treatment processes.
Microbial saccharification	Degrades polysaccharides	To obtain monosaccharides as end products it is necessary to extract the enzymes, as the sugars are intermediate compounds of the microorganisms.

Protein extraction[17] can be achieved by employing aqueous-ethanol solutions between 50 and 90% (v/v), pH variation and hydrolytic enzymes. For lipid isolation[18], thermal-mechanical processes and solvent extraction, are the most common processes. It should be also necessary to consider the environmental impact of the process and its sustainability. This will probably restrict the choice of chemicals used and the energy requirements of individual units operations.

In summary, DAFIA aims to explore two approaches for utilization of MSW: (i) direct isolation of components, and (ii) the use of the organic fraction of MSW (complete or isolated compounds) as fermentation substrate for diamines/diacids production. For both approaches, pre-treatment processes will be developed, combining biotechnological and physico-chemical processes, taking into account the natural variability of this feedstock.

DAFIA's approach for using MSW as fermentation substrate will be bench-marked against thermochemical and biological gas production, with subsequent gas-fermentation.

1.4.2 State-of-the-Art and progress in extraction-purification of nucleic acids and gelatine from fish RRM.

The main target products from MRRM are nucleic acids and gelatine. **These biomacromolecules will be obtained from fish viscera and skin/backbones, respectively.**

Nucleic acids include DNA (deoxyribonucleic acid) and RNA (ribonucleic acid), which are polymers composed of pentose sugars, phosphate and nitrogenous bases. The content of nucleic acids in fish varies among the organs. A recent study by Ravikiran et al[19] on the fresh water fish *Notopterus* revealed that tissue from liver and gonads contain twice as much DNA as kidney tissue, while the RNA content was similar in all tissues tested. The content of nucleic acids in fish milt is exceptionally high, reaching up to 50% of the dry matter. Hence, nucleic acids are currently mainly extracted from fish milt and find applications within cosmetics and nutrition[20]. Strategies for extraction and purification of nucleic acids from mixed fish viscera are poorly described in the scientific literature. Most studies use

[14] Physico-chemical processes are improvements of the established chemical processes to diminish reaction time and improve efficiency-

[15] Rabemanolontsoa H., Saka S. Various pre-treatments of lignocellulosic. Biores. Technol. 199 (2016), 83-91

[16] Sun et al.The role of pre-treatment in improving the enzymatic hydrolysis of lignocellulosic materials. Biores. Technol. 199 (2016), 49-58.

[18] Ammonia-fiber expansion (AFEX) & Ammonia recycle percolation (ARP).

[17] Villegas-Torres et al., The protein fraction from wheat-based dried distiller's grain with solubles: valorization. New Biotechnol. 32 (2015), 606-611.

[18] Abdelmoez W. and Mustafa A. Oleo. Oleochemical industry future through biotechnology. J. Oleo Science 63 (2014), 545-554.

[19] Ravikiran K., Kulkarni R.S. (2015) Int. Letters of Natural Sciences Online, 34: 1-6

[20] Rustad T (2011). Maximising the value of marine by-Products, book chapter 1, 3-18

a defined tissue, normally milt, muscle or fin, and utilize commercial kits (Qiagen, Roche etc) for isolation of nucleic acids at small laboratory scale for molecular biological purposes[21,24].

Generally, successful nucleic acid purification requires four important steps: 1) effective disruption of cells or tissue, 2) denaturation of nucleoprotein complexes, 3) inactivation of nucleases and 4) elimination of contaminating lipids/carbohydrates and proteins. Due to the complex composition of fish viscera to be used as start material in DAFIA, several strategies for nucleic acid extraction will be explored, and new methods will be developed based on their biochemical properties and on available published methods for laboratory scale[22,23], but applicable at industrial scale. The salting-out method using NaCl-saturated water for precipitation of contaminating compounds (e.g. proteins) omits the use of toxic organic solvents. Furthermore, Nasiri et al.[24] modified the salting-out method and used cheap common laundry detergent instead of expensive enzymes (e.g. Proteinase K) resulting in high yield and high quality nucleic acids. By including fish gut in the extraction process, endogenous enzymes (e.g proteases and nucleases) will be present, which together with added enzymes may be applied to facilitate the fractionation and purification process. Also the cell mass generated as a by-product from the fermentations will be utilised as a source of nucleic acids.

Fish skin and other MRRM (like backbones, viscera) are rich in collagen, and a good source for gelatine extraction.[25,29] Fish-derived collagen has been explored for a long time. Extraction of fish gelatine usually involve acid or alkaline pre-treatment of gelatine rich raw material prior to gelatine extraction[28] ·Due to a lower content of the amino acids proline and hydroxyproline, gelatines from cold-water fish species are known to have lower gel strength, as well as lower gelling and melting temperatures, compared to gelatine from mammals and warm-water fish species[26]. Fish gelatine is used in cosmetic, food and pharmaceutical applications, and in contrast to bovine gelatine, it is not associated with the risk of Bovine Spongiform Encephalopathy, and unlike porcine gelatine, it is Halal compliant. However, the suboptimal physical properties have limited the commercial interest in cold-water fish gelatine.

Another challenge is insufficient availability of raw materials and the difficulty to obtain certification on fish raw material for gelatine extraction.[29] Farming of fish (for example farming of salmon) gives uniform supply of rest raw materials whole year and guaranties quality, tractability and hygienic status of raw material. Norway alone makes almost 400 000 tonnes of salmon rest raw material,[27] which can be a start material for gelatine extraction. Different fish RRM (skin, backbones) and the insoluble fractions obtained after fractionation/hydrolysis of fish viscera into oil, stick water/hydrolysate and sediments can be used as a start material for gelatine extraction.

Optimisation of extraction from different start material like fish RRM and co-fractions after other technological processing (like thermal extraction and enzymatic hydrolysis) as well as technological steps like different extraction parameters (extraction temperature and time, pH, fractionation of extracted) as well as separation of different gelatine fractions and chemical or enzymatic modifications will be applied to improve the properties of fish gelatine.

Beyond the state of the art, DAFIA will develop cost-efficient methods for extraction and purification of nucleic acids from fish viscera, by characterisation, utilisation and adaptation of endogenous enzymes as processing tool. **DAFIA** will develop gelatine extraction technologies, improving textural characteristics so they are suitable for application as coatings. These technologies are full described in WP2.

1.4.3 State-of-the-Art and progress in microbial conversion of biomass waste to diacids and diamines.

MSW is still a highly underutilized source for microbial conversion to valuable chemicals. DAFIA project will focus on the production of the two diacids, **cis,cis-muconic acid** (CCM) and **adipic acid** (AA) (DTU and BIOPOLIS), and the **1,5-pentanediamine** (SMC), from MSW carbohydrates and the protein and lipid fractions. Strategies to handle varying MSW composition will be developed, and integrated with the strain and fermentation process development.

[21] Raja M. et al. (2011), Comparative analysis of DNA extracted from fish species preserved in formalin. Turk J Biol. 35: 331-336, [24] Triant et al., Simultaneous extraction of high-quality RNA and DNA from small tissue samples. Journal of Heredity, 100:246–250, 2009.

[22] Nishiguchi M.K. et al. DNA Isolation Procedures. Book chapter Techniques in Molecular Systematics and Evolution. 2002.

[23] Tan S. C. et al., DNA, RNA, and protein extraction: The past and the present. J Biomed Biotechnol, 2009.

[24] Nasiri H et al (2005) Modified salting-out method: high-yield, high-quality genomic DNA extraction from whole blood using laundry detergent. J. Clin. Lab.Anal. 19:229-232.

[25] Gildberg A, Arnesen JA, Carlehog M. 2002. Utilisation of cod backbone by biochemical fractionation. Process Biochemistry 38:475-480. [29] Karim AA, Bhat R. 2009. Fish gelatin: properties, challenges, and prospects as an alternative to mammalian gelatins. Food Hydrocolloids 23:563-576.

[26] Eysturskarð, J., Mechanical Properties of Gelatine Gels; Effect of Molecular Weight and Molecular Weight Distribution (Doctoral theses), in The Faculty of Natural Science and Technology 2010, Norwegian University of Science and Technology, Trondheim.

[27] Richardsen R, Nystøyl R, Strandheim G, Viken A. 2015. Analyse marint restråstoff, 2014: Analyse av tilgang og anvendelse for marint restråstoff i Norge. Trondheim: SINTEF Fisheries and Aquaculture. Report no. A 26863.

Adipic acid is currently produced from cyclohexane (93% of the global production capacity)[28] or benzene[33]. The footprint of fossil adipic acid is significant[29]. Bio-based adipic acid is estimated to bring 85% reductions in CO_2 emissions and avoids generation of N_2O. A number of companies are pursuing the development of AA by chemical or microbial catalysis from sugars or oils. Verdezyne is currently the only company that has demonstrated production and recovery of adipic acid at pilot scale (one ton/y) through fermentation of plant oil.[30] Rennovia is developing a chemical process converting sugars, including lignocellulosic, into bio-based glucaric acid, which can then be converted into adipic acid. However, the chemical catalysis processes require highly refined raw materials in order to avoid the production of side-products and to preserve the life-time of the catalyst, which will not be feasible using MRRM or MSW. Other companies that have announced to be involved in R&D on bio-based AA include DSM, Genomatica and BioAmber by microbiological routes.

Cis,cis-muconic acid (CCM) can be converted by hydrogenation into adipic acid or turned into terephthalic acid (with a global market of 71 million tons), the main component of widely used plastic polyethylene terephthalate (PET). Moreover CCM could be exploited for production of other compounds, such as diamines or mercaptoacids and esters. Several companies (Amyris/Draths Corporation, Myriant[31], Deinove[32]) have shown their interest in the biological production of CCM, but no significant progress has been announced so far. There are currently no commercial processes for bio-based production of adipic acid or CCM. The processes that are in development utilize primary food-chain sugars or vegetable oils, and not MSW as the DAFIA project proposes.

In DAFIA project we chose to work with yeast Saccharomyces cerevisiae and Yarrowia lipolytica as the microbial hosts for production of correspondingly CCM and adipic acid. These yeast are well amenable for industrial-scale fermentation, they have high tolerance to hydrolysate inhibitors and to low pH, and they have Generally Regarded As Safe (GRAS) status. Executing the fermentation at low pH has primarily two advantages: it decreases the risk of bacterial contamination and allows producing organic acids in undissociated form, which reduces the downstream processing costs. While this is of less importance for CCM, which is poorly soluble in water (~ 1g/L), for adipic acid with pKa of 4.43 and 5.41, low pH process presents a significant advantage.

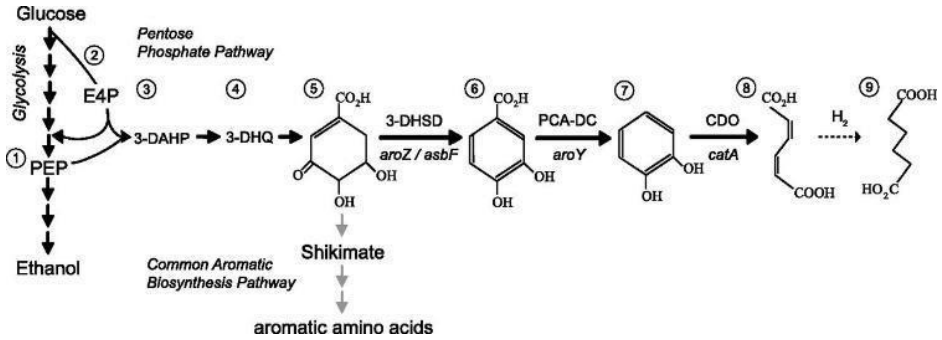


Figure 1.4.a Heterologous pathways[33] towards biosynthesis of CCM as proposed by Frost and Draths[34].

Dr. Borodina's research group at DTU has engineered *S. cerevisiae* for production of CCM at 0.6 g/L from glucose, which is the highest titter reported so far[35] (Figure 1.4.a). They also contributed to development of CCM-biosensor in collaboration with Jay Keasling's group[36]. This biosensor allows rapid high-throughput screening of CCMproducing strains using fluorescence-activated cell sorting (FACS) and thus will speed-up the strain development in DAFIA project. Via participation in another EU project BioREFINE-2G, DTU has developed robust industrial *S. cerevisiae* strain for efficient utilization of C5 sugars. These strains will serve as the platforms for creating strains for production of CCM from MSW, which contains C5 xylose, in DAFIA project. DTU has also developed a versatile genetic toolbox for *S. cerevisiae* and *Y. lipolytica*, which will allow rapid strain development

[28] www.ihs.com/products/chemical/technology/pep/bio-based-adipic-acid.aspx [33] www.groenegrondstoffen.nl/downloads/Boekjes/16GreenBuildingblocks.

[29] Van Duuren et al. "A limited LCA of bio-adipic acid: Manufacturing the nylon-6,6 precursor adipic acid using the benzoic acid degradation pathway from different feedstocks", Biotechnology and Bioengineering 108(6): 1298-1306, 2011.

[30] http://verdezyne.com/

[31] www.stapgef.org/stap/wp-content/uploads/2013/03/Commercializing-Bio-based-Chemicals.pdf

[32] www.datapressepremium.com/rmdiff/1458/PR_Deinove_Muconic_2G_final.pdf

[33] Weber et al.,"Biosynthesis of *cis,cis*-muconic acid and its aromatic precursors, catechol and protocatechuic acid, from renewable feedstocks by *Saccharomyces cerevisiae*". *Appl Environ Microbiol*, 78(23): 8421-8430, 2012.

[34] http://www.google.com/patents/US5487987

[35] Stovicek et al., "CRISPR- enables fast and simple genome editing of industrial *Saccharomyces cerevisiae* strains". *Metab Eng Commun*, 2:13-22, 2015.

[36] Stovicek V, Borja GM, Forster J, & Borodina I, "EasyClone 2.0: Expanded toolkit of integrative vectors for stable gene expression in industrial *Saccharomyces cerevisiae* strains". *J Ind Microbiol Biotechnol*, 42(11):1519-1531, 2015.

in this project[37,38]. Since both yeasts have good tolerance to low pH, the diacids can be directly produced instead of their salts, thus enabling a simpler, more sustainable and cost effective recovery process.

During DAFIA project we will create the first yeast strains capable of producing CCM and adipic acid from MSW carbohydrate feedstocks, thus making a significant advance of the state-of-the-art. First, the strains will be engineered and evolved via laboratory evolution for efficient utilization of the complex nutrients in MSW hydrolysates. Second, the pathways towards CCM and adipic acid will be implemented and the flux towards the product will be optimized via metabolic engineering[39]. The highest reported titter of CCM produced by yeast from glucose was 141 mg/L and the carbon yield was 0.9%. There are no published reports on production of adipic acid by yeast from sugars. In DAFIA project we will aim for proof-of-concept yeast fermentation process with minimum titter of 5 g/L of dicarboxylic acids and minimum carbon yield of 5%, when using MSW hydrolysate.

An alternative approach targeting adipic acid is the use of the thermophilic bacterium *Thermobifida fusca*. It has a native adipate pathway[40]. It is a major degrader of plant cell walls in heated organic materials. Nevertheless the yield and titter of the adipic acid production in *T. fusca* is still low [2.23 g/L from 50 g/L of glucose, and 0.045 g/g glucose], and has never been attempted with MSW hydrolysate. The genome of *T. fusca* _YX is available[41], as well as the metabolic model and it is possible to use state of the art molecular tools for its modification[42].

1,5-pentanediamine is currently mainly manufactured by oil-based chemical synthesis, but microbial conversion of sugars, molasses, methanol and other carbon sources into 1,5-pentanediamine has been demonstrated recent years. 1,5-pentanediamine can be produced by a single step decarboxylation of L-lysine, one of the largest biotechnological bulk products (2 million tonnes in 2015), either from sugars using engineered, lysine-producing microbes, or enzymatically, starting with lysine. Large global producers of L-lysine like Ajinomoto, Cathay and Evonik Industries AG, and have patented production processes. The most prominent L-lysine and 1,5-pentanediamine producing host is currently Corynebacterium glutamicum, which for decades has been subject for classical mutagenesis and genetic engineering aiming for increased production levels of various amino acids or the ability to metabolize alternative carbon sources, e.g xylose and methanol.

Strains of *C. glutamicum* overproducing L-lysine are available from commercial strain collections, academic groups and at SMC from previous research on L-lysine production by *C. glutamicum* and closely related *Brevibacterium* strains (confidential SMC reports). L-lysine overproducing C. glutamicum strains have successfully been converted to 1,5-pentanediamine producers through the heterologous overexpression of lysine decarboxylase genes, especially the cadA and ldcC genes from *Escherichia coli*. Secretion of 1,5-pentanediamine to the extracellular medium can be limiting for high production. However, overexpression of 1,5-pentanediamine exporters cadB from *E. coli* and a permease identified in *C. glutamicum* have proved valuable to further increase 1,5-pentanediamine production levels. Although that the technologies for engineering of bacterial strains for 1,5-pentanediamine production are known, the highest reported yields on glucose in the open literature are still only around 10 %, while the yield for lysine is 5060 %. To increase the yields and obtain competitive productivities will be a challenge in DAFIA, using MSW as a feedstock, containing mixed substrates in varying ratios and possible inhibiting compounds, and will also affect the strain selection strategies. Additional engineering related to the substrate utilisation and adaptation to the complex growth medium may be required.

The processes for the fermentative production of diacids and diamines will be developed in bench-scale bioreactors and successively scaled up to 1,500 L. Although this is much smaller than real production scale, the robustness of the process can already be tested and used to perform a realistic economic feasibility assessment. The initial development at small scale will take into account various parameters that have an impact on yield, titer and production rate. Downstream processing (DSP) will be developed and further elaborated together by BBEPP & BIOTREND for the cost-effective purification of the diamine and the two diacids. Optimization of the medium and pre-treatment process conditions will be integrated, optimized to facilitate the downstream processing on larger scale.

Beyond the state of the art, DAFIA aims at developing strains and fermentation processes to produce at least 5 g/kg (> 2 g/l) of cis,cis muconic or adipic acids, and at least 20 g/kg (>10 g/l) of 1,5-pentanediamine from MSW, carbohydrates, lipids and protein. Particularly, at least 70% of the protein and lipid fractions in MSW will be

[37] Jensen, Method for iterative chromosomal integration of multiple genes in *Saccharomyces cerevisiae*, *FEMS Yeast Res*, 14, 238-48, 2014.

[38] Jessop-Fabre MM, Jakočiūnas T[§], Stovicek V, Dai Z, Jensen MK, Keasling J, Borodina I. "EasyClone-MarkerFree: A vector toolkit for marker-less integration of genes into *Saccharomyces cerevisiae*". *Biotechnology J, in press.*

[39] Borodina I & Nielsen J (2014) "Advances in metabolic engineering of yeast *Saccharomyces cerevisiae* for production of chemicals". Biotechnol J, 9(5):609-620. doi: 10.1002/biot.201300445.

[40] Deng Y."Production of adipic acid by the native-ocurring pathway in *Thermobifida fusca* B6". *J. Appl. Microbiol.* 119: 1057-63 (2015).

[41] http://www.genome.jp/kegg-bin/show_organism?org=tfu and https://www.ebi.ac.uk/biomodels-main/BMID000000141167

[42] Deng Y. *et al.* Metabolic engineering of a laboratory-evolved *Thermobifida Fusca* mcC strain for malic acid production on cellulose and minimal treated lignocellulosic biomass. *Biotechnol Prog.* 32: 14-20 (2016).

converted. Producer strains will be designed to be robust in order to tackle variations in feedstock composition with the aim to increase the yields and to obtain competitive productivities.

1.4.4 State-of-the-Art and progress in biomacromolecules for flame retardant (FR) applications.

In recent investigations, proteins and nucleic acids have shown flame retardant properties when applied as coatings on textiles[43], well suited as these applications do not require flame retardants with thermal resistance up to 180ºC. During BRIGIT project, DLABS and AIMPLAS developed a new type of flame retardant based on the same chemistry proposed for DAFIA project, i.e., the incorporation of nitrogen and phosphorus groups to increase the thermal resistance and fireproof efficiency in biopolyesters.

The mode of action of these substances is connected to their chemical composition. For instance, the nucleic acid contains the three ingredients typical of an intumescent formulation, i.e. the phosphate groups able to produce phosphoric acid, the deoxyribose units acting as a char carbon source and blowing agents, and the nitrogen-containing bases (adenine, guanine, cytosine and thymine) that may release ammonia. Hence, its potential for flame retardant applications has been studied and demonstrated[44]. In other cases, the biomacromolecules were blended with other natural materials, such as clays and crab shells, to improve flame retardancy (as synergistic systems)[50].

The required purity of flame retardant purposes is nucleic acids mixture (A_{260}/A_{280}) >1.6[45]. Nucleic acid as a natural product requires chemical modification in order to improve its thermal and mechanical properties: indeed, as far as the former requirement is concerned, in their pristine form after extraction, Nucleic acid and proteins can be compounded only with low melting polymers, like polyolefins[46]. Therefore, the possibility of improving their thermal stability is a key factor for making the biomacromolecules suitable as efficient flame retardants for thermoplastic matrices, including polyamides. In this context, DLABS proposes in the following chemistries based on their experience in the BRIGIT project:

- Lignodulfonates, proteins and other amines can graft (or insert between) nucleic acid strands. Additional phosphoric acid or phosphate esters can be connected by acid base reactions.
- Covalent and/or ionic bond and/or by making common salts with multivalent metal cations and carbonates, silicates, borates and anions) can give synergistic effects and improve compatibility between the new flame retardant and the polyamide.

Beyond the state of the art, DAFIA aims to develop a new range of cost competitive and thermally stabilized nucleic based flame retardants, using non polluant chemical routes (free of organic solvents, reactions at temperature lower than 100ºC, etc.). These will be used in fire-protection of thermoplastics such as novel, bio-based polyamides as a real and effective alternative to ammonium polyphosphate. The thermal requirement will be a weight loss < 2% at 300ºC and not significantly decrease of the mechanical behaviour of the flame- retarded polymer.

1.4.5 State-of-the-Art and progress in gelatine from MRRM for edible/barrier coatings in food packages.

As seen, gelatine refers to the purified and modified collagen, the principal protein component in animal connective tissues such as skins, cartilages and bone. Gelatine is widely used in food, pharmaceutical, cosmetic, and photographic applications because of its unique functional and technological properties. Porcine and bovine sources account for the main production but fish gelatine has been highlighted as a better alternative to mammalian gelatines. All fish gelatines exhibit very good film-forming characteristics that lead to the formation of transparent, nearly colourless and highly flexible films[47]. Cold-water fish gelatine films was tested as barrier coating which could have relevance as edible coating on food or for bioplastic derived packaging[48].Moreover, gelatine films from warm-water fish species have been reported to exhibit stress and elongation at break similar to that of bovine gelatine. However, fish gelatine films exhibits lower water vapour permeability (WVP) than bovine or porcine gelatine, due to the lower proline and hydroxyproline content, specifically, films made from cold-water fish gelatine.

Different approaches for barrier properties improvement of gelatine films were attempted. A higher amount of denaturated gelatine led to an improvement in gas and water barrier[49]. The influence of such crosslinkers as

[43] F. Carosio, et al. "Green DNA-based flame retardant coatings assembled through Layer by Layer" in Polymer, 54 (2013), 5148-5153.
[44] G. Malucelli et al., "Biomacromolecules as novel green flame retardant systems for textiles" in: RSC Advances, 4/86 (2014), 46024-39.
[50] All-Natural Mixture Yields Promising Fire Retardant: www.nist.gov/el/fire_research/fire-060314.cfm
[45] Absorbance ratio for DNA purity assessment (UV-Vis spectroscopy)
[46] J. Alongi et al., Bulk or surface treatments of EVA copolymers with DNA: investigation on the FR properties. In: European Polymer J., 51, 112-119, 1014.
[47] Avena-Bustillos et al., Water Vapor Permeability of Mammalian and Fish Gelatine Films, Journal of Food Science, 71, 202–207, 2006.
[48] Avena-Bustillos et al., Oxygen Permeability, Mechanical Properties of Mammalian and Fish Gelatine Films. J. Food Science, 76, 2011
[49] Arvanitoyannis et al. Edible films made from gelatine, soluble starch and polyols .3. Food Chemistry, 60, 593-604, 1997.

formaldehyde and transglutaminase was studied, as well as chemical and enzymatic modifications[50]. Moreover, both water vapour permeability and oxygen permeability turned out to be affected by corn oil content and pH value[57]. Based on previous experience at IRIS using protein-based barrier coating, an oxygen transmission rate (OTR) exceeding those of all existing bioplastics and approaching that of synthetic barrier films could be reached (fig. 1.4.b, Wheylayer project).

The project is at semi-industrial level (TRL7) and also demonstrated that such barrier coatings improved the recyclability of multilayer packaging[51][52].DAFIA would provide an alternative based on MRRM proteins to previously used whey which has a high market value. Such coatings can also be applied directly on the food products as edible coatings leading to enhanced shelf life as demonstrated by previous results of IRIS researchers on frozen salmons. Enriching gelatine films with antioxidants and/or antimicrobial substances can improve their functional properties and provide an active packaging biomaterial[53]. Lignin resulted in a noticeable plasticizing effect and improved gelatine films light barrier properties, which could be of interest for preventing ultraviolet-induced lipid oxidation in certain food products.

In addition, gelatine-based films additivated with plants essential oils at low concentrations (2000-6000 ppm) exhibited antimicrobial and antioxidant properties against well-known food pathogens, without altering significantly their functional properties[54]. Moreover, other essential oils from clove and garlic were resulting in decrease of water solubility but increased the free radical scavenging activity of films. In the European project OliPHA[55], gelatine-based active coatings were developed by IRIS using a polyphenolic extract recovered from olive oil mill waste waters. The application of the coating on polyethylene or polyhydroxyalcanoate films improved by up to 5 times the oxygen barrier and provided the film with significant antioxidant activity against lipooxidation of cheese and extended significantly meat shelf life.

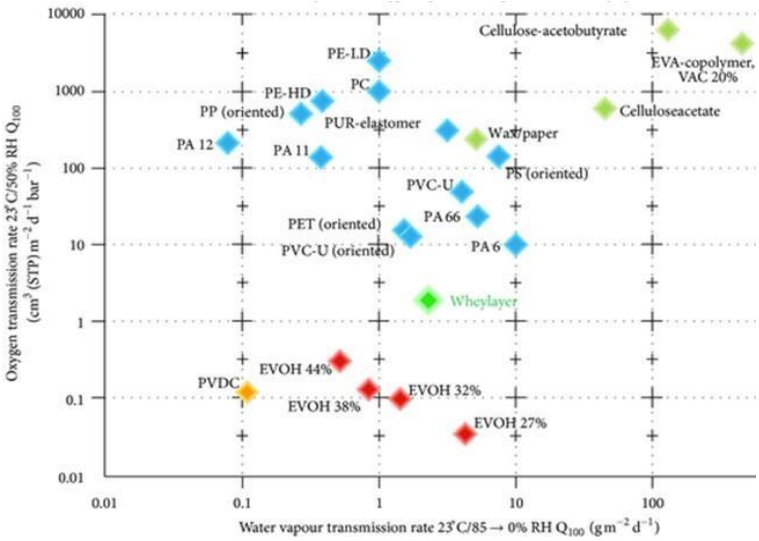


Figure 1.4.b. Barrier properties of whey-based layer versus other plastics commonly used in food packaging industry (normalized to 100 µm).

In a close loop approach, fish-derived active ingredients can also be isolated from fish muscle protein after enzymatic hydrolysis. These bioactive peptides exhibit unique structural properties and amino acid composition that have a number of promising uses such as antioxidants, antihypertensive, immunomodulatory or as antimicrobial peptides.

Beyond the state of the art, DAFIA will study the use of gelatine as barrier and/or edible coatings to improve the oxygen barrier and antimicrobial capacity of food and packaging. They will be furthermore improved by the encapsulation of active ingredients such as MRRM derived bioactive peptides which could be by electrospinning gelatine nanofibers for a more controlled release. The developed coated and/ or packed salmon fish slices will be studied in terms of shelf life with expected improved resistance against lipo-oxidation, water loss and enzymatic spoilage mechanisms.

Gelatine is a well-known food additive as well as already approved to be used in food contact materials with FCM substance nº547 (without mention to origin) with no specific migration assigned. As will be extended in the risks table, all used salmon MRRM from NUTRIMAR are suitable for food consumption. In addition, past development of biopolymers from animal by-products at IRIS, like whey protein, could benefit from exemption from being listed

[50] De Carvalho et al., Gelatin based films modified with transglutaminase, glyoxal and formaldehyde. Food Hydrocolloids, 18, 717-726, 2004.
[57] Wang et al., Effect of pH and corn oil on the properties of gelatin-based biopolymer films, J.Food Engineering 90:11-19, 2009.
[51] Processing and Validation of Whey-Protein-Coated Films and Laminates at Semi-Industrial Scale as Novel Recyclable Food Packaging Materials with Excellent Barrier Properties"; E. Bugnicourt et al.; Advances in Materials Science and Engineering, vol. 2013, Article ID
[52] , 10 pages, 2013
[53] Nuñez-Flores et al., Physical and functional characterization of active fish gelatine films incorporated with lignin. Food Hydrocolloids, 30, 163-172. 2013.
[54] Hosseini et al., Bioactive fish gelatin/chitosan nanoparticles composite films with antimicrobial properties, Food Chem. 194:1266-74, 2016.
[55] FP7 GA 280604, details in IRIS' profiles and www.olipha.eu

in Regulation 10/2011 under the scientific demonstration that the substance is safe (Article 19 says that for substances not listed in Annex I, the compliance with article 3 of the Regulation 1935/2004 is assessed in accordance with internationally recognized scientific principles on risk assessment) as long as it is pure and food grade.

1.4.6 State-of-the-Art and progress in muconic acid chemistry.

The selected diacids and diamines are monomers for polyamides and polyesters. ARKEMA is currently a producer of sebacic and other amino acids and lactam. Some diamine and fatty amines are produced from diacids using an ARKEMA patented technology. ***Muconic acid*** *(figure 1.4.c)* ***does not have an established market yet***, however, its special structure will confer it a potential value in a broad range of applications. However, if adipic acid is a necessary intermediate in the conversion of muconic acid to other chemicals, the additional double bonds would be seen as a disadvantage. Nevertheless, some direct conversion processes can be devised in order to reduce the problems. For example: 1,6-hexanediol is already produced by hydrogenation of adipic acid, and has a market in polyesters. Muconic acid can be directly hydrogenated to 1,6hexanediol. Then, it is still a single step hydrogenation, at higher pressure than the simple muconic acid hydrogenation to adipic acid. A faster solution for implementation would be to find appropriate catalyst and conditions to do that hydrogenation in existing reactors.

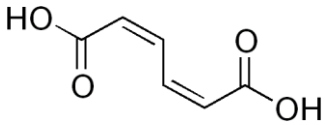


Figure 1.4.c. cis,cis-Muconic acid

Muconic acid, preferably obtained as a diammonium salt, could be converted to the diamide, and then catalytically to the corresponding dinitrile, in a similar process to what is used to make the dinitrile of sebacic acid, reducing downstream separation (DSP) costs (i.e. avoiding the intermediate conversion of muconic acid into adipic acid through hydrogenation, since anyway the nitrile has to be hydrogenated at higher pressure to the same HexaMethyleneDiAmine – HMDA). This is especially important since DSP often accounts fora major share of the final product cost. After further hydrogenation (similarly to the adiponitrile hydrogenation process[56]) it would lead to the diamine HMDA. This type of reaction is also currently done with polyunsaturated fatty acids and the losses to side products are limited. It should be noted that in this case the dinitrile would be unsaturated, therefore some side reactions may occur and the reaction would be more exothermic (than the hydrogenation of the saturated dinitrile – Adiponitrile). It will be necessary to find appropriate catalyst and reaction conditions. Since the value for HMDA is higher (than Adipic acid), and that the initial step would avoid the separation/extraction there are significant economic advantages. Also the hydrogenation of the unsaturated dinitrile would not add much in capital and operating conditions compared to the hydrogenation of the saturated hexanedinitrile/adiponitrile (fossil-based). The best option is to hydrogenate at the end (the unsaturated nitrile), rather than first hydrogenate muconic acid to adipic acid, then make the nitrile and then make the second hydrogenation to the HMDA. It is even more important if the fermentation produces an ammonium salt of the diacids, in order to reduce the number of steps.

Muconic acid bearing two unsaturations can be converted to a thiol-diacid by reacting H_2S on the double bonds. This reaction is known for maleic acid derivatives (mercaptosuccinate[63]) and for the dimercaptosuccinic acid derivative. Mercaptosuccinic acid-based polymers have been described in polymers and adhesives formulations[57]. So it is expected that sulphur derivatives of Muconic acid would have similar or improved properties. Although the **proof of concept** has been done for the reaction with maleic acid derivatives, it still has to be demonstrated that the same reaction conditions can be used for industrial grade muconic acid, and that the sulfur derivative has improved properties in the final application. Other synthesis processes for mercaptoacids and esters are also known in the literature and will be considered[58-59]. What is planned is to use it as an additive to improve the adhesion of polyamides on metal surfaces. Such an application has been very recently patented with more sophisticated monomer synthesis[60]. That means that a part of the adipic acid in the formulation would be replaced by a part of the sulfur compound. So the structure would remain a polyamide. The technical advantage is expected in the final application when several layers of products need to be used (i.e. when a first adhesive binding layer is coated and then a polyamide or polyester adding the number of operations) and where a single adhesive layer of the new polyamide containing the muconic derivative could be used. In this scenario the product value would be based on the displacement of a more complex and expensive market incumbent. . Its value could then be much higher than the simple adipic acid, especially if the effect can be obtained at low dosage (less than 10 wt. %) in the polymer formulation. Value of additives in these type

[56] Z. Jia et al., Liquid phase hydrogenation of adiponitrile over directly reduced Ni/SiO2 catalyst, Catalysis Communications, 73, 2016, 80-83
[63] Production of mercapto compounds, US Patent 5,008,432 to Phillips Petroleum Company (1991).
[57] Reversible crosslinked polymers, WO Patent 2013/163655 to WYSIS Technology Foundation.
[58] Process for the synthesis of 3-mercaptopropionic acid esters, US Patent 5,919,971 to Elf Aquitaine Exploration (now part of Arkema).
[59] Method for synthesising a mercaptan by adding hydrogen sulphide to an olefin, WO 2015059412 A1, 2015.
[60] hwww.xanchem.com/ and Polymer Having Adherence Properties WO 2016/020297 assigned to Xanchem, 2016.

of formulation can reach more than 10 €/kg. For the upscaling, ARKEMA is used to handle sulfur compounds, since ARKEMA is a leader in sulfur chemistry.

In summary, DAFIA aims to use renewable muconic acid as an additive for adhesion promoters in polyamides, and as a source for HMDA, by develop new routes to thiol derivative synthesis of diacids.

1.4.7 State-of-the-Art and progress in renewable monomers for bio-based polyamides.

At present, in production of commercially available bio-based PA, castor and palm oils are used as a renewable feedstock. 11-aminoundecanoic acid and sebacic acid are products from chemical conversion of ricinoleic acid, the major fatty acid component of castor oil[61]. The modifications may lead to synthesis of polyamides building blocks, such as sebacic acid, aminoundecanoic acid or decamethylenediamine, which are schematically represented in Figure 1.4.d. Moreover, some polyamides can be produced from sugars. For example, PA 6 could be synthesized by ringopening polymerization of □-caprolactam produced by glucose fermentation. However, production of bio-based PA from carbohydrates is not cost-effective yet. Figure 1.4.e shows a scheme of some of commercial bio-based polyamides and their percentage on renewable source and main properties. The properties of polyamides increase with increasing the length of the monomer chain. Long-chain PAs have lower density, lower water absorption but also significantly lower strength and stiffness, lower melting temperature and continuous operating temperature than standard polyamides. Few companies have already introduced different kinds of long-chain bio-based polyamides into the market.

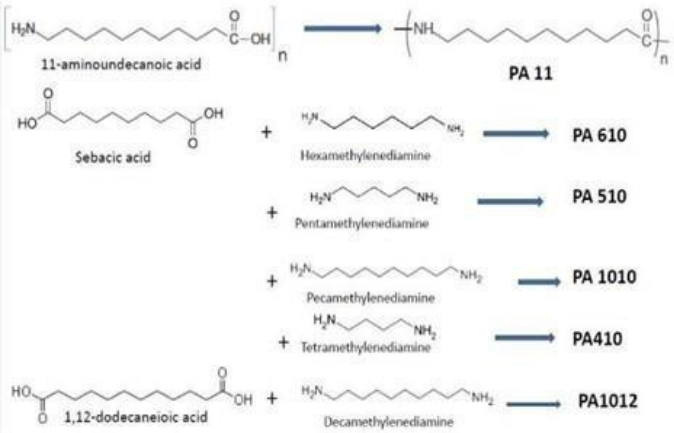


Figure 1.4.d Schematic route of bio-polyamides production from natural oils (castor and palm kernel oil).

Polyamide 11 (Rilsan from Arkema) has been commercially available and applied in different industry sectors (e.g.: automotive, textile, electronics), mainly in coatings.

Polyamide 10.10 (offered by DuPont, EMS-Chemie, Evonik and Suzhou Hipro (now Arkema)) has properties close to PA11. Both PA 11 and PA 10.10 are fully made from castor oil. Polyamide 5.10 is now produced by BASF, but not yet commercially available.

Its dicarboxylic acid part is synthesized from castor oil, while diamine is obtained by glucose fermentation, via lysine as intermediate[62], see section 1.4.4. Polyamide 6.10 (BASF, DuPont, EMS, Rhodia, Evonik, Toray, Suzhou Hipro (Arkema), Akro-Plastic) with about 60 % of renewable-source content has recently met with a great interest, as it combines high mechanical properties (characteristic of PA 6) with low density and high chemical resistance (typical of PA 10.10). Other partially biobased polyamides available on the market are PA4.10 (up to 70 % biobased, DSM Company) and PA 10.12 (about 40 % biobased, Evonik, Suzhou Hipro (Arkema))[63]. Chinese company Cathay Industrial Biotech Ltd has also a provided long-chain diacids and 1,5-pentane-diamine to the polymer industry and Ajinomoto and Toray have a partnership on Biobased polyamides – for the C5 diamine[71] from the amino acid lysine produced from plant materials. Biopolyamides 5.6 has the same strength and heat resistance as conventional polyamide made from the petrochemical derivative HMDA. However, its current percentage of biobased source is 45% (according Cathay's company).

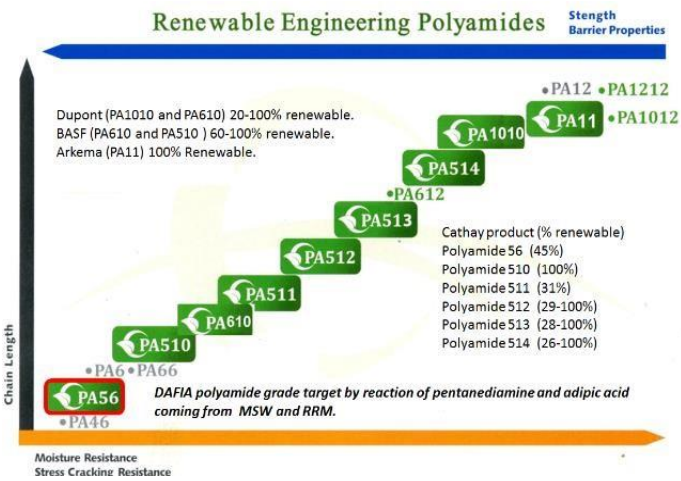


Figure 1.4.e. Main commercial available biopolyamides (source : modified figure from Cathay webpage).

[61] S. Kuciel et al., Polyamides from renewable sources as matrices of short fiber reinforced biocomposites, *Polimery*, 9, 627-634, 2012.
[62] Kind S, Wittmann C. Bio-based production of the platform chemical 1,5-diaminopentane. Appl Microbiol Biotechnol. 2011;91:1287–96.
[63] Kolb N et al., Long-chain polyesters and polyamides from biochemically derived fatty acids. Eur Polym J. 2014;51:159–66. [71] www.cathaybiotech.com/en/ and www.ajinomoto.com/en/presscenter/press/detail/g2012_02_13.html.

Beyond the state of the art, DAFIA aims to use diacids and diamines (conning from RRM and MSW) as a raw material for building blocks and intermediates in the production of different polyamides (mainly PA5.6) with a percentage of biobased sourced raw materials close to 100%.

2. IMPACT

2.1 EXPECTED IMPACTS

Table 2.1.a Contribution to the expected impacts defined in the *BIOTEC 02:2016* topic text.

Expected Impact in the topic	Expected contribution of the project.
Develop at least 2 new bio-based value chains, utilising either industrial by-products and/or relevant bio-waste fractions for bio-product generation.	Three *bio-based value chains will be developed*: i) Flame retardants from nucleic acid, ii) coatings from gelatine for food packaging applications; iii) dicarboxylic acids and diamines from fermentative process of carbohydrates, lipids and proteins. Two bio-wastes are selected: municipal solid & fish waste.
Propose outline business plans that include the assessment of the potential impact of the proposed value chains in terms of EU jobs and growths in the short and medium term.	DAFIA plans to look for alternatives to petroleum based chemicals with large worldwide markets (Flame retardants, coatings and polyamides) and also to explore new market/applications for chemicals from renewable resources chemicals based on improved functionalities (such as adhesion promoters of polyamides by modifying muconic acid with H_2S). A preliminary business model has been prepared by the different partners based on their current sales and consumptions, the estimated target cost, their current market share and the expected market evolution. This aims to have a turnover of more than 200 M€/year and creates 670 direct employees. NNFCC partner is company specialized in the preparation of business plans for the biotechnology companies to put in the market new bio-based products with a great succesfull. However, due to the TRL at the end of the project, we expect a significant sales 5 years after project finish when TRL 8 will be achieved. More accurate calculations of profit is difficult due to the current low yield of some proposed routes. Expected costs are determinated based on the expertise of the DAFIA's partners and the methodology proposed by Hermann [64] to calculate the technoeconomic prospects of new routes for bio-based bulk chemicals. Furthermore, the validation of the cell design competences and technologies which will be developed and deployed during DAFIA can be translated in advanced microbial cell engineering services and green chemical routes whith high potential of generating highly qualified employment in the short term.
Demonstrate the environmental feasibility of the proposed value chains and conduct relevant outreach activities.	A full LCA will be developed during the project for each of the three bio-based value chains (Flame retardants, edible and barrier coatings, and different biopolyamides). Selected wastes are not used for human consumption. A high percentage of selected wastes are not properly valorised. A full dissemination and communication plan is presented later in this section. All project partners have a broad experience in outreach activities from previous EU projects.
Activities are expected in TRL 3 to 5. The participation of SMEs is encouraged.	DAFIA targets TRL 3 – 5 as table 1.3.a explains and involves seven SMEs. The project consortium includes **8 SMEs** with a significant role in the project. They represent 52.46% of the person month effort and 46.80% of the budget.

2.1.1 Enhancing innovation capacity and integration of new knowledge.

Valorising municipal solid waste (MSW) and marine rest raw material (MRRM) sectors has an environmental rationale and creates an important competitiveness opportunity for Europe as described in the European Commission strategy for 'Innovating for Sustainable Growth: a bioeconomy for Europe'. Indeed, addressing our societal challenges is a major catalyst for increasing our innovation capacity and for the integration of new knowledge generated as a result of research into practical societal and industrial contexts.

The percentage of chemical production based on biotechnology is estimated to increase from less than 2% in 2005 to approximately a quarter of all chemical production[65] by 2025. In EU28 2013, the chemical and plastics represented a turnover of **42 billion €** and 200,000 employees[66]. Initiatives such as DAFIA is an enabler for innovation in countless

[64] Herman B.G. and Patel M. "Today's and Tomorrow's bio-based bulk chemicals from white biotechnology. A techno-economic analysis". *Appl. Biochem. Biotechnol.* 136: 361-388 (2007).
[65] Smeets, E. et al. (2014) Evaluating the macroeconomic impacts of bio-based applications in the EU.
[66] http://biconsortium.eu/sites/biconsortium.eu/files/news-image/16-03-02-Bioeconomy-in-figures.pdf

other sectors that can transform these wastes into sustainably sourced materials raw materials as inputs for countless discreet manufacturing and process industries in Europe and internationally.

To increase the innovation capacity of the industry, our approach is to capitalise on a valuable opportunity for the EU fisheries and municipal waste management, chemicals, plastics and other end-user industries to work together in order to substitute currently used fossil-based counterparts with bio-based materials and to assume a leadership position on the world stage.

The innovation process is only complete when new innovative knowledge is put to industrial use. The innovative potential of DAFIA is enhanced by the **strong industrial composition of the DAFIA consortium (10 partners out of 16; 8 of them SMEs and two large companies**). Industrial partners are integrated into the project decision making structure to ensure that the involved industrial sectors drive project activities and benefit from the project results. To ensure success, industrial partners will lead all the technological WPs and enable the steering of project activities from the perspective of the whole supply and value chain.

DAFIA's success will be mainly based on the strong interaction and partnership of SMEs/industry and research organizations, all highly skilled in complementary fields as figure 2.1.a shows.

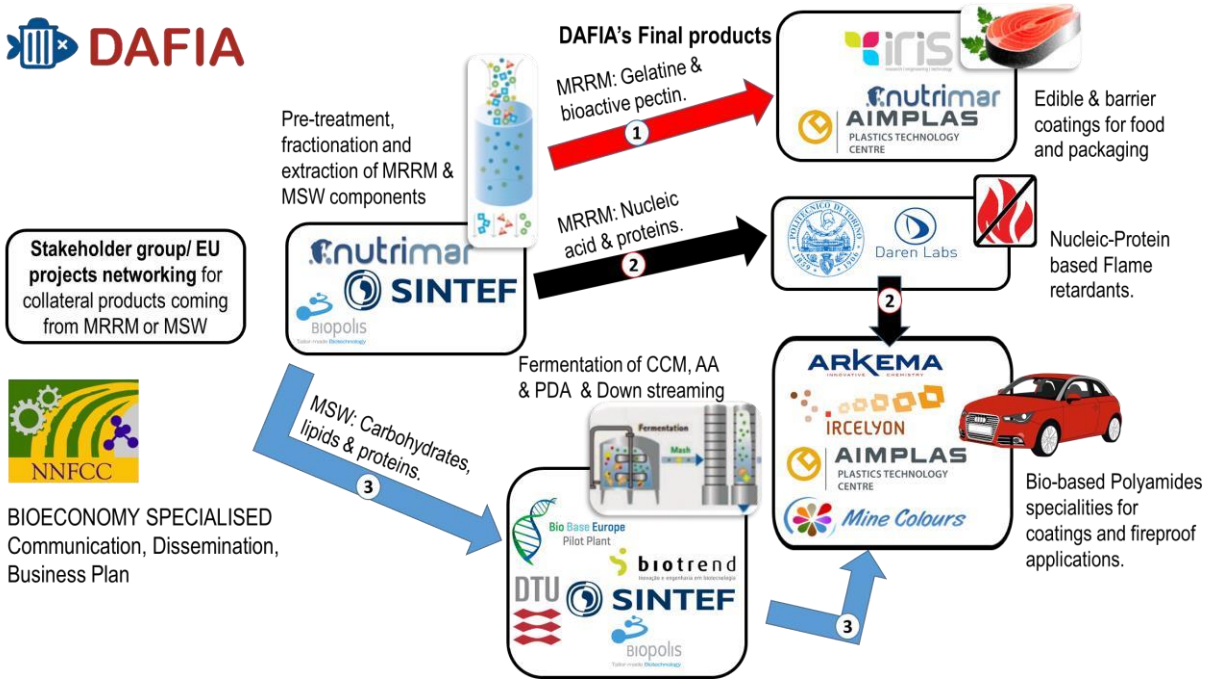


Figure 2.1.a. Structure and relation between partners in the consortium.

Three main routes are defined in DAFIA project: ① NUTRIMAR, SFA and SMC will be in charge of extraction and purification of gelatine and other bioactive additives from MRRM, in a second stage IRIS will modify the extracted/purified products to produce an edible and coating barrier, the edible barrier will be validated by NUTRIMAR and some of their customer, the coating barrier will be validated by AIMPLAS. ②Norway's partners will be in charge of extraction and purification of nucleic acid and proteins from MRRM, DLABs and POLITO will be in charge of chemical modifications of these MRRM components to produce high efficient flame retardants that can be used to modify polyamides or other non-polar thermoplastics.③ BIOPOLIS and SFA will be in charge of pretreatment of MSW to be suitable for fermentation of selected diacids and diamines, after their fermentation carried out by DTU, BIOPOLIS and SMC, the monomers will be purified by BIOTREND and BBEU, that will be in charge of the scale-up process. The monomers will be chemically modified to produce polyamides (PA 56) and intermediates (in combination with SH2) to produce polyamides (and potentially polyesters) with improved adhesion properties to metals and glass. The polyamides will also modified by the flame retardant of route 2 to be used in applications where fire could be produced easily. These polyamides have a broad range of applications in sectors such as cars, EE, sport, pipes, etc.

Cross-fertilization and synergies among partners with expertise in complementary and cascaded processes (IRIS, BIOTREND and BBEU), technologies and products will act as a booster for innovation placing better and more sustainable products on the market. The participation of large end-users guarantees high impact and the wide dissemination & exploitation of the project results at worldwide level.

2.1.2 Economic impact: strengthening the competitiveness and growth of companies

High economic impact will be achieved through strengthening the competitiveness of the Industrial Partners in the target market sectors and related ones (flame retardant, intermediated (diacids and diamines) and polyamides and edible and barrier coatings for food and packaging) by: 1) promoting the use of DAFIA materials, components and systems in high value-added applications while keeping costs competitive, 2) strengthening and differentiating the EU Industry, 3) increasing workers' skills. Detailed information of each market is presented:

Flame Retardants (FR).The worldwide consumption of ***flame retardants (FR)*** amounts to about 1.8 million tons p.a.[67] and is expected to rise to 2.4 million tons by 2018. Approximately 85% of all FRs are used in plastics uses[68]. FR market value is projected to reach 9.5€ billion by 2019[69], growing at an annual rate of 6.1% till 2020. Aluminium hydroxide (ATH) was the most widely used FR in the world in 2013, accounting for nearly one-third of global consumption, and it is expected to retain its dominance in the product mix through 2018.However, further growth is difficult as high dosages near 60% w/w are required to be effective, significantly reducing the mechanical properties of the final product. Due to regulatory issues, it is expected that phosphorus-based FRs will record the fastest market growth while halogenated FRs , such as bromine- and chlorine-based products, are being phased out[78]. As a consequence, the global market for non-halogenated flame retardants is expected to reach 2 million tons by 2020, growing at an estimated CAGR of 7.3% in terms of volume. Phosphorous based FRs are expected to generate over 50% of the overall market revenue, growing at an estimated CAGR of 8.3% 2014 till 2020. This is estimated to be the fastest growing segment in revenue terms. However, Phosphate Rock has been listed in the Critical Raw Materials[70], alternatives reducing Phosphate dependence are necessary and are an important objective for DAFIA project.

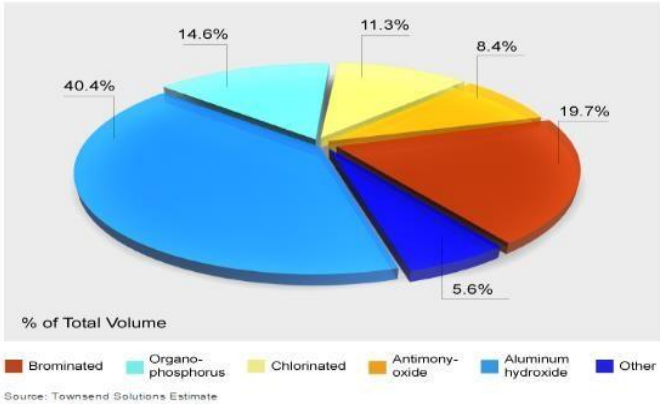


Figure 2.1.b Global Consumption of FRs in Plastic by type,2011

It is expected that DAFIA's ***new FR additive based on the use of nucleic acid and proteins will replace Ammonium polyphosphate (APP) in many of its current applications and using similar dosage.*** The market possibilities are therefore large, currently, approximately 300,000 tonnes of APP is produced globally every year by large companies[71] such as Clariant, Albemarle, Budenheim, Unibrom with a price of 5-7€/kg. Taking 5€/kg as a median price for nucleic acid-based FRs, this results in a total potential market of more than 1,500M€. This initial cost has been calculated based on the results and the chemical routes employed in the BRIGIT project as follows;

As DAFIA's FRs contains nucleic acid and protein from viscera and fermentation microorganism and lignosulfonates (a by-product of the wood industry), we do not expect that the cost of the raw materials at industrial scale will be beyond 2.5 €/kg, the cost of chemical modifications were estimated as the 70% of raw material cost, other costs including sales and marketing cost, overheads and profit margin will represent 30% of material cost.

Competitors and key stakeholders in the flame retardant market. An analysis of the structure of the market players and key stakeholders demonstrates that the market is not controlled by large players and not limited by the required raw materials. Identifying and effectively negotiating technology licensees will be a key activity for DAFIA partners. The product performance and sustainability demands of final consumers will provide a strong market pull.

Existing Competition: DAFIA Flame retardant (FR) offer a proprietary, innovative, eco-friendly and effective alternative over existing commercial products. As no product can offer a bio-based eco-friendly flame retardant, DAFIA will carve out a market position in the use of FRs mainly in polyamide applications. Furthermore products would be sold at a lower price than existing solutions and they, as in the case of APP, can be used in other polymers such as biopolyesters, hence significantly contributing to increase the project impact.

Potential New Entrants: As the flame retardant market is growing, with an increasing number of applications, it is inevitable that there will be additional companies interested in developing and selling bio-based FRs. However, it is anticipated that the majority of uptake will be from existing manufacturers who will be interested in licensing the

[67] www.bccresearch.com/market-research/chemicals/flame-retardant-chemicals-chm014l.html
[68] www.flameretardants-online.com/web/en/106/7ae3d32234954e28e661e506e284da7f.htm
[69] Markets and Markets (2014) Flame Retardant Market by Type (and End-User Industry)- Global Trends & Forecast to 2019
[78]www.marketwatch.com/story/world-flame-retardants-market-2015 and www.marketsandmarkets.com/plastic-additives.asp
[70] http://ec.europa.eu/growth/sectors/raw-materials/specific-interest/critical/index_en.htm
[71] //polymer-additives.specialchem.com/additives-flame-retardants-smoke-suppressants-ammonium-polyphosphates-app

patented DAFIA technology. By allowing the technology to be licensed, DLABS and AIMPLAS will reduce the incentives for these companies to undertake the costly, slow and risky process of developing alternatives.

Materials Suppliers: the main ingredients of DAFIA-FRs will be modified proteins, nucleic acid and lignosulfonate, by-products of the fish industry and the acid digestion of wood. Both are available worldwide at very low price since they are very abundant. Other chemicals required for the modification of the nucleic acid are used in several processes and are available in the required quantities to meet market demand, with established prices and several suppliers.

Main customers: The main customers of flame retardants are large polymer manufacturers such as ARKEMA and small or medium size compounders (that produce tailor made plastic for specific applications) such as MINE. ARKEMA and MINE will offer their own customers new polyamide grades with green FRs at a competitive cost.

Political Groups and Consumers: There are many consumer groups looking for actions to reduce and regulate the use of halogenated FRs, and several directives taken by governments across Europe and the World also aim to achieve this. To date there has been a trade-off between the long-term harmful effects of the halogenated FRs and their effectiveness at slowing the spread of fire and thereby saving lives. DAFIA will offer the opportunity to provide biobased FRs that are both effective and safe. Effectively disseminating this information to will be essential to drive a non-governmental organizations (NGO) and consumer pull for DAFIA-FRs.

Intermediates (diacids and diamines) and Polyamides (PA) market.

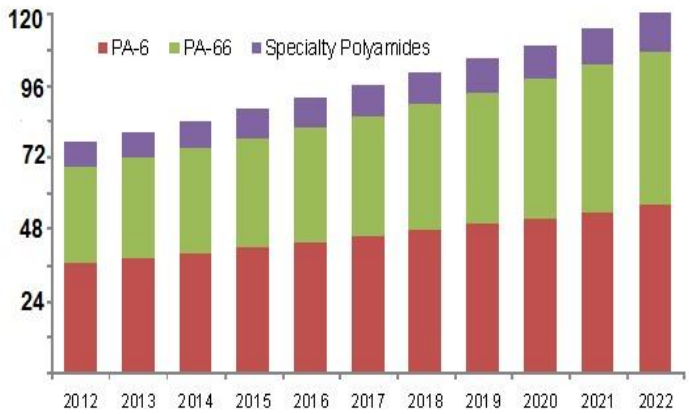
Adipic acid is a white, crystalline compound and is primarily used in the production of polyamide 6.6. It is also used to produce polyurethanes, as a reactant to form plasticizers, lubricant components and polyester polyols. Feedstocks include benzene, cyclohexane (CX) and ammonia. Key end-use products include automotive textiles, shoes and shoe soles. Other outlets are as a food ingredient in gelatines, desserts and other foods that require acidulation. According to ICIS Consulting, between 2012 and 2018, European consumption is forecast to increase from 705,000 tonnes/year to 742,000 tonnes/year, whereas the production capacity is estimated in 850 kt/year in five European plants operated by BASF, LANXESS, SOLVAY and RADICI. The price of dicarboxylic acids or diamines produced from industrial sugars depends of the length of the chain, but it is in between 1.5-3 €/kg.

Muconic acid (MA), a high value-added bio-product with reactive dicarboxylic groups and conjugated double bonds, has gathered increasing interest owing to its potential applications in the manufacture of new functional resins, bioplastics, food additives, agrochemicals, and pharmaceuticals. Muconic acid is currently a speciality chemical, its market size is limited by its cost of production. According to a market report[72], the global muconic acid market was worth 46 M€ in 2013 and will reach an estimated value of 75 M€ by the end of 2020. Geographically, North America and Europe will continue to dominate the global muconic acid market.

Some of the top market players that supply industry with muconic acid derivatives are USA companies: Santa Cruz Biotech, and Sigma-Aldrich. The growth of the muconic acid market will be driven by the development of new market applications and enabled by increasing consumer preference towards bio-based products[73].

The growth of the muconic acid market will also depend on the ability to either, reduce the product price by reducing the cost of raw materials, making process improvements or a combination of both; or through the development of **new high value applications reliant on the chemical properties offered specifically by muconic acid**.

In consequence, DAFIA will propose **the use of muconic acid in applications where their double bond will be an advantage to produce high added value chemicals** (as an intermediates and adhesion promoters in polyamides and polyesters see section 1.4.5). Value of additives in these types of formulations can reach more than 10 euros/kg. Based on the market knowledge of ARKEMA, muconic acid derivates represent a market of 10,000t/y. The muconic acid will be mainly used for modification of speciality polyamides using in coating applications. The **1.5 diaminopentane** is a sought-after platform chemical used for production of various polyamides and is currently mainly fabricated by petroleum-based chemical synthesis. Bio-based diamine is produced from 100% plant sugars using a patented process developed in Cathay Biotech's and by Ajinomoto in collaboration with Toray. The polyamide market is projected to reach 38 Billion € by 2020, at a CAGR of 5.4% from 2015 to 2020. This growth is majorly driven



[72] www.transparencymarketresearch.com/muconic-acid-market.html"Muconic Acid Market- Global Industry Analysis, Size, Share, Growth, 2014-20"

[73] NNFCC Bio based Products Market Review. 2015.

by the increasing demand from the automotive as well as packaging & textiles application segments. The biobased and specialty segment is expected to witness high growth from 2015 and 2020, as end user industries are switching to this type of polyamides for enhanced performance in specific applications. The current market of polyamide 56 and other bio-based polyamides[75] is very low (less than 70,000 t/y) in comparison with petrol based (near 8 Mt/y) as shown in figure 2.1.c, however their potential is quite high since they could be a real alternative to PA6 or PA6.6, both polyamides have a worldwide market of 7Mt worth 18 Billion €. Specialty or long-chains polyamides are being used in many industrials applications. These polyamides (PA6.10; PA6.12; PA10.10; PA11 & PA12) bring more technical performance in terms of flexibility, thermal resistance, moisture resistance, stress cracking resistance and polar fluid resistance. Price is very dependent on performance and end-use application, but it is between 10-15 €/kg with a potential demand of 200 kt/y.

Figure 2.1.c *Global bio-polyamide market volume by product, 2012-2022 (Kilo Tons)*[74]

Competitors and key stakeholders in the polyamides market. An analysis of the structure of the market players and key stakeholders demonstrates that the bio-based polyamide market is controlled by large players like ARKEMA, Evonik, Solvay, Dupont and Cathay Industrial Biotech. ARKEMA has a business focus on speciality polyamides, for polyamides 5.6, AIMPLAS will identify and negotiating licensees with key industrial companies.

Existing Competition: DAFIA's Polyamides will offer significant advantages over the existing competitive solutions. As no producer can offer a 100% bio-based eco-friendly polyamide 5.6 (TERYL 5.6 of CATHAY only contains 45% of the renewable material). There are long chain polyamides from renewable sources, but the inclusion of muconic acid in the polyamides as adhesion promoters will open new market opportunities for polyamides in coatings with direct application in several markets. Furthermore, it will be competitive in cost and properties versus existing petrol based solutions, due to the use of the same chemical monomers as currently used but coming from low cost and renewable raw materials.

Potential New Entrants: As the polyamide market is growing, it is inevitable that there will be additional companies interested in developing and selling bio-based polyamides. By allowing the technology to be licensed, AIMPLAS will reduce the incentives of companies to undertake the costly, slow and risky process of developing alternatives. ARKEMA is worldwide leader in sulphur-based compounds, it is difficult to find real competitors in this field.

Materials Suppliers: the main ingredient of DAFIA bio-based PA are diacids and diamines, coming from MRRM and MSW. Raw materials are abundantly available worldwide at very low price. Additional additives required for polyamides polymerization (catalysts, solvents, chain extenders, etc…) and processing (nucleants, lubricants, antioxidants …) are used in several processes and are readily available, with established prices and several suppliers.

Main customers: The main customers of polyamides vary from large plastic processing companies or compounders (such as AIMPLAS' suppliers or ARKEMA's customers) to small and medium size processing companies and compounders such as MINE. ARKEMA and MINE will offer their customers new bio-based polyamide grades at a competitive price. Licencing agreements with other chemical companies (such as UBE & NUREL external partners) or compounders (POLYONE, KAFRIT, CLARIANT) will enable broader project impact.

Political Groups and Consumers: There are many consumer groups looking for actions to reduce and regulate the use of non-renewable source materials or renewable materials in competition with the food chain (as UNILEVER internal policy), and several directives taken by governments across Europe and the World also aim to achieve this. However, these groups face a trade-off between the reduction of the dependency from oil-based materials/CO_2 diminution and their effectiveness to preserve foods or car safety and comfort. DAFIA polyamides are both effective and safe. Effectively disseminating this information will be important to consumer pull for bio-based PA.

Fish gelatine-derived edible and barrier coatings for food and packaging market.

On the one hand, **edible coatings** are predicted to expand very significantly in the future, as health-conscious consumers look for more food products that require minimal preparation like cut fruit and premixed salads as well as food products with enhanced shelf life. The edible-film industry has grown dramatically from 10 companies the mid-1980s to more than 1,000 companies now with annual sales exceeding €85 million[76].

On the other hand, functional additives and **barrier coatings market** for plastic packaging is projected to reach €3.2bn by 2018, at a 5.7% annual growth, according to a recent report from Smithers Pira[1]. In addition, the **active & intelligent packaging market** was valued at €1.9billion in 2015 and is demonstrating a strong growth trend with, for example the US demand forecast to increase 8.0 percent annually to €3 billion in 2017 and Europe being the second biggest market behind US.

[74] http://www.grandviewresearch.com/industry-analysis/bio-polyamide-market
[75] www.european-bioplastics.org/bioplastics/materials/
[76] M.E. Embuscado, K.C. Huber (2009). Edible Films and Coatings for the Food Applications. Springer. ISBN- 978-1-4419-2830-6

Competitors and key stakeholders in the market: **Edible coating solutions** as planned in the project are not commercially available and today the main uses of edible coatings remain for fruits (waxes), sweets, cheese, etc. whereas the main planed application here is for fish using also fish components. The annual market of EVOH (Ethylene vinyl alcohol), the **main barrier material used in the flexible food packaging**, was in the range of 110 kt per year worldwide (source EVAL company). The share of the EVOH use in food packaging is 43%. Available active packaging are led by gas scavengers. Barrier coatings with antioxidant/ antimicrobial properties alike the one planned to be developed here could be applied to a large percentage of flexible packaging market.

Potential New Entrants: Several other EU-projects are incubating solutions based on different bio-based feedstocks that could provide some of the functionality planned in DAFIA. Nevertheless, none is based on MRRM or combine such a broad toolbox of active/edible/barrier coatings solutions to find the best approach on a case-to-case basis.

Materials Suppliers: Global **fish production** has grown steadily in the last five decades, **increasing at an average annual rate of 3.2%.** Worldwide fish processing industry leads to the production of a **large amount of MRRM** which is generally discarded (**~7.3 million tons/year**). As such there is an **increasing amount of MRRM available** for example for the production of gelatine and derivatives to be used in applications such as DAFIA's edible and barrier coatings for food and packaging optionally enhanced with antioxidant/antimicrobial properties with bioactive peptides from MRRM.

NUTRIMAR already produces fish protein concentrates and plans to deploy the hydrolysis process developed in WP2 in order to supply DAFIA's applications post project. Nevertheless, fish gelatine is already available worldwide at affordable price, but it is mainly obtained from meat (pig and bovine sources being the major sources) instead of from the fish skin as in the case of DAFIA. Additional additives required for barrier and edible films will be selected based on natural origin and economic viability criteria.

Cost of barrier polymers[77] such as EVOH or PVDC is 6-8€/kg, therefore in view of the low cost MRRM supply and considering processing and additives costs, it is realistically feasible for DAFIA to be cost competitive with current supply. The use of an edible coating on a salmon slice is estimated to cost less than 1 euro cent which is considered attractive to the market given the accompanying improvement in shelf life and product quality.

Main customers: The main customers of the barrier and/or active coatings are packaging converters such as films plastic processing companies. The main costumers for the edible coatings are food producers, especially fish processors, as it will be the main application tested in the project. SALMAR, largest salmon producer in EU, and supplier of MRRM to NUTRIMAR, was identified to have keen interest in both solutions.

Political Groups and Consumers: As we have pointed, there are many consumer groups looking for actions to reduce and regulate the use of non-renewable source materials. Their effectiveness to preserve foods which is where DAFIA targets to land its superior food protection performance vs. established biopolymers.

Effectively disseminating this information to these groups will be essential to drive a high level pull for bio-based packages and edible coatings. In summary, the potential impact of the products to develop in DAFIA project and the current and expected cost are summarized in the table 2.1.c. Expected costs are evaluated based on the expertise of the DAFIA's partners and the methodology proposed to systematically evaluate the techno-economic prospects of future production routes of biobased bulk chemicals by Hermann et al.[69]

Table 2.1.c Potential economic impact of DAFIA target products.

Potential applications	Current// Expected price (€/kg)	Worldwide consumption (kt/y).
Green FRs (alternative to APP).	5-8 // 4-5	300
C6 diacids (adipic acid).	1.5 // 1.5	5,000
C6 Diamine.	2 // 2	1,000
Muconic acid specialities.	10	New market (10)
Polyamides for coatings.	10-15//10-15	50
PA5.6 & fireproof polyamides.	3-5 // 4-5	7,000
Edible and barrier coatings.	6-8//4-6	30

2.1.3 Business model proposed by each partner for future commercialization of DAFIA developments.

[77] http://omnexus.specialchem.com/tech-library/article/materials-developments-for-barrier-food-packaging-applications

Regarding the **segments in the value chain directly represented in the project**, and thus project partners, the expected impact is described below, which will result in an increase of the potential market and increased high added value product portfolio of DAFIA's partners:

- **Renewable chemicals by microbiological routes (BIOTREND, SMC, DTU, BIOPOLIS, DLABS and BBEU):** will select and optimize strains to maximize the yield of the bioreactor. The combination of novel strains to obtain valuable products from complex by-products coming from extraction and separation of MRRM and MSW is the **main and innovative product to develop in the DAFIA project**. The typical business model for these partners would be to patent the modified strains, and use the validations in DAFIA to either licence out the strains and their applications or enter into joint ventures with companies which are already active in this area for the production of chemicals. Several examples exist in the industry that validate this business model: Genomatica, Metabolic Explorer, Global Bioenergies, BioAmber and Deinnove are all examples of companies that emerged as research-based start-ups after deriving novel engineered strains with differentiating traits enabling them to produce, for the first time, specific bio-based products from renewable resources. The DAFIA's partners are very active in the international industrial biotechnology market as service providers and are investing in the generation of their own IP in order to progress in the value chain and retain more value of their activities.
- **Polyamides and/or polyesters with improved adhesion** (ARKEMA) based on the use of muconic acid to develop new polymers containing adhesion promoters special designed for several industrial sectors. ARKEMA is a large company, which will use a combination of business models to introduce its new products in the market. The company expects moderate growth in these mature markets, strong growth in developing world therefore the exploitation focus will be on developing markets. Selling directly to customers and also utilisation of wellestablished existing distribution networks has proven a very effective business model in other lines of products and it is expected that these will be suitable for DAFIA products. ARKEMA does not exclude the possibility of joint ventures although this will be considered after establishing a strong market presence in other geographic regions. Production of new products will be considered within current assets or through toll manufacturing.
- **Polyamides with green flame retardants (MINE, AIMPLAS and ARKEMA):** ARKEMA and MINE aims to formulate polyamines with the flame retardants produced by licencing companies of DLABS, the compounds will be sold directly to their customers in selected countries, using the current sales network of each company. Countries not covered by both companies will be reached by licencing the technology by AIMPLAS or to provide special formulations in similar polyamides. Several compounders will be contacted. DLABs will negotiate individual licensing agreements with ARKEMA and other EU flame retardant manufacturers in order to cover all the European market in a first step and USA market in a second step.
- **Fish gelatine-derived edible and barrier coatings for food and packaging (IRIS, AIMPLAS and NUTRIMAR):** novel, low cost affordable formulations of edible and barrier coatings will be developed; the formulations will be optionally enhanced with active ingredients that will be food (contact) compliant, toxicsubstance free, flexible, increase food conservation and/or packaging recycling, cost effective, etc. IRIS plans to either patent the formulations if novelty allows it or to keep them as trade secrets that could be licensed to NUTRIMAR (expanding its line of gelatine products) and to other edible coatings manufacturers so they can provide royalties while being commercialised in the broad packaging and food markets. AIMPLAS will have an exclusive use of the coated film for their current market places. Licencing to other companies will be proposed.
- **Other related industrial sectors**: Other cost-competitive applications will be proposed and evaluated along the project. We expected that all the work done on Muconic acid could be transferable to other unsaturated diacids such as 1) Maleic or Itaconic acid, which would give mercaptosuccinic acid, 2) C18 unsaturated diacid can be obtained either by fermentation or by self-metathesis of oleic acid, 3) C11 and C12 unsaturated diacids can be obtained by cross metathesis of C10 or C11 unsaturated acid with aryclic acid/ester… Any improvement on thiol derivative synthesis of diacids will benefit other industries, for example in the production of pesticides such as Malathion. The diacids can be transformed into diesters to be employed as ~~a~~ solvents, plasticizers or lubricants.

Business models and plans will be developed to support the commercialisation of each key project result and partner. The Business Model and Plan developed in WP9 will integrate IPR, innovation protection and exploitation strategies, and the accompanying techno-economic analysis and LCA developed in WP8. This will pave the way for the successful exploitation of the project results. The expected impact, based on a conservative scenario of the value chains, is described in qualitative and quantitative terms in table 2.1.d. This results in an estimated benefit (an average 15% of turnover) of 230 M€ over the period 2025-2030 (5 years after project end). It is important to highlight that the environmental benefits have not been included in these calculations.

Table 2.1.d. *Preliminary evaluation of DAFIA economic impact after 5 years project finish (when TRL8 is achieved).*

Potential applications	Expected price (€/kg)	World Market size (kt/y) /Market share[78]	Expected Turnover[79] (M€/y)	Direct Employment	IP management	Required investment
Nucleic acid based FRs	4-5	300 kt/ 1.5% (4,500 t/y)	25-30 M€ (MINE & ARKEMA needs)	130 jobs (225k€job)	Patents & Industrial secret.	Adaptation of existing Chemical plants.
C6 diacids: Adipic acid & muconic acid.	1.5	5,000 kt/ 0.2% (10,000 t/y)	15 M€ (ARKEMA & PA56 production needs).	35 jobs (400k€//job)	Patent & licensing to biological companies.	Build a new plant or use existing one.
C6 Diamine: 1-5-pentanediamine	2.0	1,000 kt/ 0.7% (7,000 t/y)	10 M€ (PA56 production needs)	25jobs (400k€/job)		
Muconic acid-SH_2 derivatives.	10	New market 10,000 t/y	100 M€ (ARKEMA customer's needs).	250 jobs (400k€/job)	ARKEMA owns IP	Build a new plant (ARKEMA interest).
Polyamides coatings (ARKEMA).	10-15	50 kt / 2.5% (2,500 t)	35 M€ (ARKEMA sales).	85 jobs (400k€/job)		Existing polymerization units
PA56 & fireproof PAs.	2.5-5.0	7,000 kt/ 0.15% (10,000 t/y)	40 M€ (ARKEMA, MINE and licencing companies sales)	100 jobs (400k€/job)	Licencing to chemical companies	Existing polymerization units
Edible and barrier coatings.	4-6	30 kt/ 2.5% (750 t)	4 M€(AIMPLAS' customer's needs)	20 jobs (200k€/job)	Use & Licencing to coating firms.	Use of existing processing units.

Additional market research will be carried out in WP9 to assess and quantify the needs of the market: existing competitors, target markets, customers' profiles etc., this analysis will confirm and validate market opportunities and exploitation routes. One of the issues of the envisaged fermentation processes is the fact that only a fraction of the carbon will be converted into the envisaged final products and a significant amount of microbial biomass (i.e. the microbes which are grown in the fermenters) will be produced which should also have some type of valorisation at the end of the process, e.g. used in anaerobic digestion for biogas production or others, energy cogeneration, etc.

2.1.4 Environmental and socially important impacts.

On 13 February 2012, the European Commission adopted a strategy for «Innovating for Sustainable Growth: A Bioeconomy for Europe»[89]. This strategy proposes a comprehensive approach to address the ecological, environmental, energy, food supply and natural resource challenges that Europe and the world is currently facing.

With the world population set to approach an estimated 9 billion by 2050, against a background of finite natural resources, Europe needs renewable biological resources -not just for securing healthy food and animal feedstuffs but also for materials to produce several bio-based products.

Environmental impact: will be reduced because new bio-based products will be marketed using environmentally friendly technologies from the cultivation to the final production, showing a positive CO_2 balance relative to existing fossil-fuel derived products or other bio-based sources. Table 2.1.e shows the estimated differences between current (fossil based) and DAFIA approaches in terms of Global Warming Potential (kg CO_2-Eq.) and Energy Demand (MJ) per kilo of produced polymer or intermediates. Due to the use of waste by-products out of agricultural, preferential use of microbial synthesis, highly effective catalysts, low reaction temperatures, green chemistry routes, we expect a significant reduction in both parameters.

Table 2.1e DAFIA expected impact on renewable source content and energy demand.

	Oil based materials[90].	Potential reduction using DAFIA concept.

[78] Expected conservative market share (%) based on current worldwide consumption and production capacity of DAFIA partners. In brackets t/y of DAFIA products that will be produced for target market share.

[79] Only considers the turnover the amount of material that DAFIA industrial partners can produce and commercialize (ARKEMA, NUTRIMAR, & MINE) or licensing (BIOPOLIS, BIOTREND, DLABs, IRIS & BBEU).

	GWP (kg CO_2-Eq.)	Energy Demand (MJ/kg).	Renewable content (%)	GWP (kg CO2-Eq/ % less).		Energy Demand (MJ/% less).	
Diacids (C6) Adipic acid	25.1	136.0	100%	15.1	40%	122	10%
Polyamides	8.0	137.0	100%	5.6	30%	137	0%
Ammonium polyphosphate	2.1	32.0	80%	1.1	50%	27	15%
Acrylic resin coating	7.1	120.0	100%	3.6	50%	90	25%

For example, most of the adipic acid commercialised today is obtained by catalytic oxidation of a cyclohexanone and cyclohexanol mixture (using nitric acid as a catalyst), both of which are obtained from benzene. This reaction also yields nitrous oxide (N_2O), which represents an environmental concern, since N_2O has a GWP that is 298 times higher than CO_2. Most adipic acid producers have made significant investments to reduce the N_2O emissions. Nowadays, a few percentage of adipic acid can also be obtained from: (1) fermentation of fatty acids from vegetable oils (biological process) – from Verdezyne, and (2) oxidation of glucose (chemical process) – from Rennovia.

The environmental impact of adipic acid and

Table 2.1.f Environmental impact of biobased and fossil chemicals.

Impact Category	*Bio-Adipic Acid* (AD)	*Fossil AD*	*Bio Polyamides*[92] *(PA)*	*Fossil PA*
Climate Change ($kgCO_2eq$)	-1.4 to 9.2	24.7	4.0	8.0
Land Use (m^2.a)	1.1 to 7.5	0.08	11	0.0012
Non-renewable energy (MJ)	3.2 to 195.4	121	208	122

related polyamides was summarised in an EU study[91] based on LCA data for production through the fermentation of sugars using different raw materials and purification methods. Selected data is detailed in table 2.1.f, which shows that from an overall point of view, for the categories of climate change, bio-based

[89] //biobs.jrc.ec.europa.eu/sites/default/files/generated/files/policy/EU%20Bioeconomy%20Communication%20Strategy%20Brochure.pdf.
[90] a) Althaus H.-J. et al. (2007) Life Cycle Inventories of Chemicals. Final report ecoinvent data v2.0., vol 8. Swiss Centre for LCI, Empa - TSL. b) Hischier R. (2007) Life Cycle Inventories of Packaging. Final report ecoinvent data v2.0., vol. 11. Swiss Centre for LCI, Empa - TSL.
[91]//biobs.jrc.ec.europa.eu/sites/generated/documents/BISO%20EnvSust%20-%20Bioproducts%20-%20Adipic%20Acid%20v2_150728.pdf
[92]www.fkur-biobased.com/fileadmin/user_upload/Produkte/Vestamid_Terra/VESTAMID_Terra_-_Life_Cycle_Analysis.pdf

derivatives possess environmental benefits when

compared with their fossil counterparts.However, bio-based products often show lower environmental performance for impacts categories such as eutrophication, acidification and land use. The higher impacts found for bio-based products in these categories are a consequence of the agricultural activities (such as the use of fertilizers, pesticides and energy) necessary for the biomass production.

In DAFIA will expect higher GWP reduction as the raw material is a true waste, currently without any valuable use, and doesn't need land for its production. Extraction and purifications will represent the main impacts.

Furthermore, the adoption of safe-by-design approaches will support EU competitiveness in this particular research and innovation field. Regulators can additionally use the well-defined predictive tools that underlie the concept of safe-by-design and thus improve the regulatory preparedness by timely interactions with innovators developing novel applications of polymers and intermediates coming from MSW and MRRM by-products. These interactions focus on sharing views and expertise on how best to deal with new safety aspects of innovative products for workers, consumers and the environment. A LCA of the key exploitable products will be undertaken in WP8 led by AIMPLAS, with the support of commercial software and the real data obtained at lab and pilot plant level.

Employment impact:

DAFIA aims at a significant contribution in the large number of jobs that green chemistry will create in the coming years. We expect at least **670 qualified jobs directly** related to the sale of new bio-based products five years after the end of the project (when TRL8 will be achieved), based on a very conservative European market share of 1-3% as table 2.1.d detailed. According the CEFIC, the sector generated an even greater number of indirect and induced jobs – up to three times higher than through direct employment, i.e., **DAFIA can create up to 2,000 of indirect jobs.**

In order to remain competitive and maintain jobs in the light of major societal challenges and rising markets in the developing world, the European bioeconomy sectors need to innovate and further diversify. It is estimated that direct research funding associated to the Bioeconomy Strategy under Horizon 2020 could generate about 130,000 jobs and € 45 billion in value added in bioeconomy sectors by 2025.[80]

In 2014, according to Cefic[81], the European chemical industry produces 16.7% of the world's chemicals, employs 1.2 million workers and contributes €527 billion to the EU economy. While the value of chemical production grew about 45% from 1997 to 2017, direct employment in the EU chemical industry decreased by an average annual rate of 1.9% from 1996 to 2013. The number of employees in 2013 was 28% less than the 1997 level. i.e., a loss of more than 400,000 jobs. That works out at a 1% drop in jobs for every 1.6% increase in output of non-pharmaceutical chemicals. Jobs have fallen, because sales have not kept up with increases in productivity that requires fewer workers. Along with jobs created by gaining new customers and being on the leading edge of greener chemicals, additional jobs will be created by switching from chemical feedstocks to renewable raw materials. Renewable raw materials including wastes and residues will require local processing creating EU jobs and displacing imports of petrochemical raw materials. For example a US report states[95] that if around 20% of chemical production switched from making petrochemical-based products to plant-based products that would create/transfer 104,000 jobs.

2.1.5 Policy framework

DAFIA will contribute to reducing the EU's dependence on fossil fuels and imported biomass and contribute to meeting its energy and climate change policy targets for 2020. To achieve this, **DAFIA** contributes to the two essential elements identified by the **EU BioEconomy strategy**: a) by promoting the full valorisation of waste biomass for the production of resource efficient and sustainable bio-based products through biorefinery, enabling the substitution of fossil-based products and reducing biomass dependency in regions where the biomass availability is less plentiful[82] and b) by providing sound Dissemination and Business plans to support the commercialization of DAFIA bio-based products.

More generally, DAFIA fully aligns and supports the **EU Innovation Policy and the EU 2020 strategy**, as the outcome of DAFIA will have significant impact for the municipalities, fish processors, waste managers, biorefineries, downstream processors and bio-based product producers and all industrial application sectors with the potential of using bio-based products, contributing to the **EU Strategy for Key Enabling technologies**[83] by advancing the knowledge in biotechnologies in order to provide advanced materials. High economic impact will be achieved through increasing the competitive advantage of EU Industry by promoting the use of DAFIA bio-based products in

[80] BECOTEPS (2011) The European Bioeconomy in 2030: Delivering Sustainable Growth by addressing the Grand Societal Challenges
[81] //fr.zone-secure.net/13451/106811/publication/contents/templates/Cefic_F_and_F_2014-Full_report_Blanc_150dpi.pdf
[95] www.greenbiz.com/news/2011/05/12/green-chemistry-holds-potential-reverse-job-losses
[82] http://www.crossborderbioenergy.eu/gis.html#6
[83] "A European strategy for Key Enabling Technologies – A bridge to growth and jobs ", COM(2012) 341 final

high value-added applications. It will also consolidate the EU Industry and reduce the EU dependency on crude oil. Regarding the **Lead Market Initiative (LMI) on bio-based products**, DAFIA will produce a number of high valueadded bio-based products for several demanding sectors. All these products have been identified in the LMI Final Report[98] as being of high strategic impact and social interest. DAFIA technical approach, focused on sustainable and cost-competitive biotechnologies, and all the accompanying measures planned in the project to foster the market introduction of such bio-based products will have a major impact in the EU, protecting the environment and human health and strengthening industrial competitiveness. DAFIA will provide substantial elements that will contribute to the majority of the Priority Recommendations provided by the **LMI** Ad-hoc Advisory Group for Bio-based Products[1]. NNFFC's partner was a member of this group and current members of the Commissions Expert group for bio-based products. In addition, the integration of the biotechnologies and methodologies developed in DAFIA into current industrial processes is fully in line with the Environmental Technology Action Plan.

Finally, DAFIA aims to cover most of the key objectives of the **Strategic Innovation and Research Agenda (SIRA)** published in 2013 by the **Bio-Based Industries consortium**, related to the development of chemical products from biomass by-products, including complex biomass as MSW or animals by-products. DAFIA partners such as ARKEMA, BIOPOIS and BBEU are full members and AIMPLAS and SINTEF are associated members. IRIS, ARKEMA and SINTEF are SPIRE members.

2.1.6 Barriers.

The barriers which may affect the expected impacts of the project, and the proposed mitigation measures to remove these barriers, are presented in the table 2.1.g.

Table 2.1.g Main barriers identified and the proposed mitigation actions.

	BARRIER DESCRIPTION	MITIGATION ACTIONS
Technical	Technical risks inherent to an R&D project, 1)developed bio-based products do not allow for the commercially competitive productivities 2)High downstream process costs 3) High process energy costs. 4) Low quality resins and intermediate products in comparison with petrochemical based ones.	A risk and contingency plan has been outlined and risk monitoring strategies (see section 3.2) have been implemented in the work plan (see WP10, Task 10.2 Risk indicators definition and risk management).
Market	Initial Small Market Size for MSW and MRRM based products. : Low demand can lead to high unit costs and does not allow volumes to grow enough to lower the price (no economies of scale). Brand resistance to the use of waste derived materials in food packaging applications. European policy and support for bio-based products and bioeconomy development is unclear.	Different market products will be proposed, with huge potential demand and medium cost such as coatings, packages, polyamides, flame retardants and also chemical intermediates which are present in a several industrial products. In view of the compatibility for food contact and/or food application of selected materials, the brand owner may turn the use of food processing by-products as a marketing claim as opposed to food-competing crops used in many bioplastics production. Through proactive communication DAFIA will provide information to policymakers on project results reinforcing the value of bioeconomy development.
Social acceptance	Imperfect Flow of Existing Information about green chemistry safety: In ideal markets, buyers and sellers are fully informed on the features of the good or service under consideration; however, existing information often is not equally available to all parties. Lack of information about service life and durability.	**DAFIA** will provide well documented results, dissemination materials and product datasheets including characterization and environmental data, etc., that will form a sound base to support high level development of regulations and green procurement policies. During the project extensive testing campaigns will be carried out to assess performance and long-term durability and realistic estimations will be provided by the main project players.

Industrial	There is a risk that the processes and technologies developed in DAFIA cannot be scaled up at acceptable costs and that no economies of scale are generated when industrializing the processes.	Transfer of DAFIA innovations to industrial partners during the project to test and validate the scalability of the processes. Economic projections of the full-scale production.
Standards	Need to be fulfil the current standards. Since October 2011, CEN, CENELEC and ETSI are working towards the creation of a	**DAFIA** Consortium will follow closely the outcomes of the active WGs and whenever a project outcome is considered to be of relevance to the on-going work, **DAFIA** Partners will

[98] //ec.europa.eu/enterprise/policies/innovation/policy/lead-market-initiative/files/final-eval-lmi_en.pdf

	BARRIER DESCRIPTION	MITIGATION ACTIONS
	solid standardization framework for bio-based products, based on the outcomes of CEN/TR 16208[99]. The EC issued four standardisation mandates for bio-based products **ISO/TC 276** and **CEN/TC 411, 183 & 249/WG 17.** More information in section 4 (Annex 4.2).	establish a liaison (NNFCC is part of the UK standards group for bio-based products & AIMPLAS is already participating in standardization) to present project results and if possible to contribute to the standards (e.g. CEN workshop agreements).
Regulation	Current legislation is concerned by the potential health, safety and environmental risks in relation to bio-based products obtained from biorefineries	**DAFIA** project will review the legislation applicable to this technology in its different areas: i) Regulations REACH and CPL (classification, labelling and packaging of substances and mixtures) and Legislative Decree EU 10/2011, ii) Directive 91/689 defining hazardous waste (update: 2008/98/EC), iii) Decree 52/97 defining hazardous substance: Assessment of the risk to the health and safety of workers from the presence of chemical agents, iii) Environmental impact assessment (EIA), strategic impact assessment (SIA), air and waste sectors, Legislative Decree 152/2006 and Decision 2000/532/EC, etc, to ensure compliance of the developed products

2.2 MEASURES TO MAXIMISE IMPACT

Communication, dissemination and demonstration activities are critical for the successful exploitation of DAFIA research and innovation findings and accordingly a whole work package will be dedicated to these actions (WP9). Communication, dissemination and exploitation are linked activities and must be undertaken in a coordinated way, thus exploiting synergies and avoiding overlap between them; in DAFIA they will be coordinated by a **Dissemination and Exploitation Board** (**IPREB**, see section 3.2. for a detailed description). Activities and their impacts will be **continuously monitored** during the project and the resulting actions will be reported in all Contractual progress reports. Actions will be reported with subsequent plans for the next project period in the interim & final **Plan for Exploitation of the Results and Dissemination (PERD)**, due in months 18, 36 & 48.

DAFIA will implement a range of activities to ensure the optimal visibility of the project and its results, increasing the likelihood of market uptake of the knowledge it produces, and ensuring a smooth handling of the individual intellectual property rights of the involved partners to pave the way to knowledge transfer and exploitation. Table 2.2.a summarizes the focus of activities and how the dissemination and exploitation plan fit together to produce the overall PERD.

Table 2.2.a. Plan for exploitation of the results and dissemination (PERD)

Communication activities	Purpose – Raise awareness of project aims and outputs amongst broad stakeholder base. • Communication KPIs to measure effectiveness and efficiency. • Messaging and visual identity of the project and Stakeholder mapping. • Communication channels (website, social media, printed materials, scientific & trade journals)

Dissemination plan	Purpose – Support project exploitation through technical dissemination and industrial outreach. • Dissemination strategy: objectives, targets, activities and channels. • Dissemination KPIs to measure the dissemination effectiveness and efficiency. • Management of the three dissemination activities: – packaging knowledge for an effective take up, – reaching the selected early adopters to motivate for changes, – preparing the effective exploitation of the project results. • Dissemination administration (approval, reporting, deliverable). • Scientific and trade publications.
Exploitation plan	• IPR strategy and Exploitation plan for the project results • Business models and plans for key project results

2.2.1 Communication activities

Communication activities will use a range of communication channels to spread new knowledge and results generated by the project. They aim at reaching the dissemination objectives and reinforcing the plausibility of the exploitation

[99] www.cen.eu/cen/Sectors/Sectors/Biobased/Pages/documents.aspx

plan. In order to carry out the project outreach in a comprehensive way, a detailed communication plan will be implemented at the beginning of the project (month 6). It will take the form of a full-fledged communication campaign to be designed and rolled out in parallel with the other project activities. It will aim to maximize the impact and visibility of the project among all relevant stakeholders in the fermentation, coatings and chemical manufacturing sectors at European level.

The communication leader will coordinate the upcoming activities and follow-up on the implementation of the campaign. Any successful campaign is based on a thorough, well-structured strategy. This entails the:

- Identification of the target audiences that should be reached (WHO).
- Tailoring of the messages towards these target audiences (WHAT).
- Decision on the best communication channels and tools to reach the target audiences (HOW).
- Development of a detailed implementation plan (WHEN).

The clear **categorisation of target groups** allows dedicated and targeted dissemination actions to be undertaken through specific communication channels and using targeted messaging.

Our communication plan encompasses a comprehensive series of actions (Table 2.2.b) to **broadcast the nonconfidential results that are generated during the project to as wide and relevant an audience as possible** in order to heighten awareness about the results, to contribute to knowledge building in Europe, and to ensure that as many actors and citizens as possible have access to these results. This includes the broader European chemical, food and plastic industry, policy makers, researchers and the academic community, training institutes, early stage researchers, mass media (TV, radio and the written press) and multipliers for addressing the general public, as well as other stakeholders, such as other H2020, EC and national R&D initiatives, ESIF stakeholders, etc. In keeping with the general profile of each target audience, the most appropriate communication tools and channels will be selected for targeting each stakeholder group.

Table 2.2.b Outline of the DAFIA Communication Plan

Target Audience	Tools & Channels	Expected Impacts

MSW plant management and fish industry. Food packers and packaging convertors Producers of coatings and their customers. Producers of plastics parts (Automotive, EE, …) Environmental consultancies & engineering Machinery Retailers and consumers Industry Multipliers- Sectorial Associations (e.g. EuPC, EuropaBio, CEFIC, European Bioplastics, etc.) Standardisation bodies and policy stakeholders, national food safety authorities, EFSA, EC, national governmental bodies, ICC, AACCI, CEN, ISO, WHO, FAO, CODEX, AENOR, General public and Mass media Research and academic stakeholders Early stage researchers Government & policy makers Innovation programs Educational stakeholders and	Project corporate identity tools (logo, tag line/slogan) Project Website, Project Leaflets, Project Posters, Powerpoint Presentations, Press releases Video clip (for YouTube, project website,...) ***Trade fairs/exhibitions***: K fair, FAKUMA and Interpack (Ge), Hispack (Es) ***Conferences, seminars and workshops*** (participation in organising, scientific committees, with poster, paper, presentation, or chairing): Annual World Food and Innovation Forum; European Bioplastics, Biopolpack, Biopol, EFIB, etc. ***Mass media & Magazines***: Newspapers, local TV, Packaging today; Plastic additives, bioplastics magazine, etc. **Open Access Journal Articles**: J. of Food Engineering; J. of Environmental Monitoring and Assessment; Packaging technology, J. applied Polymer Sci., etc. ***Involvement in networks/clusters***: Green chemistry, European Bioplastics, EuropaBio, Suschem, Trade Associations, etc. ***Features in Newsletters and e-bulletins*** ***Visibility on third party web sites*** e.g. LinkedIn, Slideshare,, Pinterest etc ***Inclusion in EU dissemination resources*** such as the EU Horizon Magazine and the EIP Agri Website	Generation of awareness about DAFIA and its mission to ensuring sustainable use of selected bio-waste in Europe and globally. Dissemination of new knowledge and technologies for transitioning polymer related industries to the green economy. Effective knowledge sharing among the research and scientific communities, which will contribute to the ERA, and to the EU recovery package and the EU 2020 strategy for smart, sustainable and inclusive growth. Sound communication towards policymakers to increase the influence of the project and its results in policymaking (which will be reinforced by the presence of our expert Associations). Communication to retailers and consumers about the efforts being made by the MSW and Fish sector and the target products sectors to increase their sustainability for consumer and environmental protection. The general public should also feel they are getting value for money from public funded research that will ultimately benefit them as citizens. Attract more young people to scientific and research careers, enriching their curricula with knowledge from the

Target Audience	Tools & Channels	Expected Impacts
students Other cross-cutting H2020 projects and international/ national projects PPPs and EIPs (e.g. EIP Raw Materials, BIOTECH, SPIRE, BBI, Suschem, etc.)	//ec.europe.eu/eip/agriculture and EIP raw material //ec.europa.eu/eip-raw-materials/ Social media: twitter, facebook, utube, LinkedIn, Google+ etc… Programme meetings Policy roundtables	project. Early stage researchers inspired by the results, will be motivated to remain in EU- research and possibly work in areas that are beneficial for society, sustainability and innovation.

2.2.2 Dissemination

The main communication activities are **focused on the dissemination of technical information to industry**. The goal is to disseminate **DAFIA** objectives and results among the interested industry sectors, demonstrating the added value, innovation and new technologies developed within the project. Due to the relevance of this target audience, the position of the project consortium within the industry is very important.

Project results will disseminated through participation in international conferences and trade fairs, and through publications in peer reviewed scientific journals and trade publications in the area of biochemistry, polymer processing, food waste processing, bioplastics, etc. Publication in **Scientific Conferences** related to the **DAFIA** R&D areas will be also issued. Table 2.2.c details the number of publications and the journals where the consortium plans to contribute. The project will disseminate the results to business stakeholders to make them aware of the expected impact of the project and promote the exploitation of its results. Presentations shall be made by the partners at business seminars and trade fairs. Additionally, each of the partners will disseminate **DAFIA** results in the fairs and exhibition they participate.

Table 2.2.c Activities related to participation in international events and publications in technical journals.

DAFIA's partner	Conferences & seminars.	Publications in high impact journals
AIMPLAS/ DLABS/ ARKEMA/ POLITO/ MINE	Seven contributions as speaker or poster in green chemistry, polymers and fire.	Six in Journals devoted to Fire science, polymer science polymer degradation or green chemistry.
DTU/BIOTREND BIOPOLIS/SMC	8-10 contributions as speaker or poster in conferences or fairs related to biotechnology.	Six in e.g., Metabolic Engineering (IF=6.7), Biotechnology for Biofuels, Microbial Cell Factories, Biotechnology and Bioengineering, applied microbiology or similar.
CNRS/ ARKEMA	3-4 contributions as speaker or poster in conferences related to green chemistry.	4-5 in peer-reviewed journals such as Journal of Catalysis, Topics in catalysis, Applied Catalysis, ACS catalysis,…
IRIS/NUTRIMAR SFA/AIMPLAS	2-3 contributions in conferences related to food and food packaging as speaker	Five review paper in Journal of food science Process Biochemistry, J. Aquatic Food Product Technology, Biotechnology Reports.
NNFCC	4 conferences to promote the project findings and engage in outreach activities.	None envisioned

Open Access will be provided to peer-reviewed papers published by DAFIA's Partners as table 2.2.c indicates, either in an open access journal repository ("gold" model) or for instance using the OpenAire orphan repository. Depending on the results published, the selection of the best suited and higher impact repositories will be made on a case by case basis in cooperation with **OpenAIRE**. Provided that the IPREB do not identify issues conflicting with the industrial exploitation of project developments. Technical press platforms, generic or specific will be continuously used, as well as the own partners communication channels.

Networking networks: In addition to the direct commercial contacts of DAFIA industrial partners, wider dissemination would be achieved through using the partner membership of trade associations related to the aims of the project, working with trade associations the consortium would increase the potential for exploitation of project results and additionally create a mutual opportunity with the associations to promote green and sustainable chemistry within the relevant industrial sectors. Moreover, the know-how transfer of **DAFIA** to the European industry could

be also provided by all DAFIA partners via existing contacts to other European research institutes and networks, such as European Bioplastic, BBI, CEFIC, EuPC, SPIRE, ECP4 and EuPC.

Training and knowledge transfer: DAFIA will work to enhance the training of designers, plastics application technicians and education of young polymer engineers. In particular, the ***Guide to explore the conversion routes of municipal solid waste (MSW), and marine rest raw-materials (MRRM), to obtain high added value products*** (Task 9.6 and D9.8) delivered by the project will be a very useful tool to achieve this goal. This guideline will be published by the consortium and will include new important design rules for the various application areas, including the case studies of the demonstrator parts with all the necessary information for the adaptation of results to new product developments. As the guide will be published at the end of DAFIA, technical and economical evaluations for the developed prototypes will be included. This guideline will be a crucial tool for the potential end-users of DAFIA materials and technologies. DAFIA aims to Organise 2 workshops (anticipated at month 36 and month 48). The topic will be selected a year ahead of the workshop, depending on the results obtained in DAFIA across technical, economic and environmental areas. DAFIA will showcase the demonstrators developed during the project. The Partnership will analyse the feasibility to collocate the workshop to a major fair/exhibition in order to maximise the impact.

2.2.3 Exploitation of results.

The exploitation routes and corresponding business models will be investigated in WP9. The route for commercialization of the results can be logically identified from the **DAFIA** WP structure and work plan, from the logical flow of the project activities and their clear role, as represented schematically in **figure 2.1.a**.

Exploitation expert NNFCC will act as coordinator of the exploitation activities and ensure that all IP-owning partners as well as potential and confirmed exploitation partners will work towards a successful technology and knowledge transfer to ensure efficient market uptake. The role of the exploitation manager will be to assess the commercial potential of the results and liaise with industrial companies to maximize their commercial use. The exploitation manager will follow the progress made in DAFIA and will develop full exploitation plans as the unique values of the project results are confirmed. The exploitation manager, with support of the members of the IPREB composed of representatives of all partners, and will also be responsible for establishing targeted exploitation and dissemination strategies, when the project outcomes reach sufficient maturity. The entire project group will discuss this strategy and the corresponding document will be delivered for the mid-term assessment (D9.3-M18 and 36). An updated and more detailed exploitation plan will be submitted thereafter (D9.6-M48).

All technologies developed by the partners will be considered for exploitation, be it methods, processes or materials. A summary of anticipated ***Key exploitable results (KER)*** is reported in table 2.2.d. Exploitation issues will be brought to the attention of all partners at an early stage in the project to ensure that issues regarding market and commercialisation, identified during exploitation analysis, can be fed into the R&D actions of the project.

Table 2.2.d Summary of the key exploitable results.

WP	Key Exploitable Results (KER)	Partner Interested In Exploitation
1, 2	Develop methods for extraction and purification of nucleic acids from RRM (viscera) and from the microbial biomass.	SFA, SMC, NUTRIMAR & POLITO.
2	Technologies for pre-treatment, hydrolysis and fractionation of MSW, for use as a fermentation substrate.	BIOPOLIS, SMC and DTU.
2, 5	Develop of gelatine extraction technologies, improving characteristics to employ them in the production of coatings (yields: 35 % in fish skin).	NUTRIMAR & SFA
3	Microbial hosts design for production of diacids and diamines, from the carbohydrates, protein & lipid fractions of MSW or RRM.	DTU, SMC & BIOPOLIS.
3, 7	Develop a robust fermentation processes, tolerating variations in feedstock composition, providing efficient & high productivities of the target products.	DTU, SMC, BIOTREND & BBEU.
3, 7	Design novel and environmentally friendly processes for the recovery and purification of diacids and diamine from complex fermentation media.	BIOTREND & BBEU.
4, 7	Chemically modify selected polymeric by-products for use as flame retardants	DLABS, ARKEMA & AIMPLAS.
5, 7	To develop fish gelatine-based active, edible and/or barrier coatings for application on food and packaging.	NUTRIMAR, AIMPLAS & IRIS.
6,7	Bio-polyamides flame retardant compounds suitable for automotive industry.	AIMPLAS, ARKEMA & MINE.

6,7	Production of esters, nitriles, alcohol or diols, thiols from muconic acid as monomers for polyamides and polyesters.	ARKEMA
6,7	Bio-based polyamides 5.6 from adipic acid and 1.5-pentanediamine.	AIMPLAS

The first phase of exploitation activities will aim to raise awareness among the project partners on exploitation issues In the first 6 months, NNFCC will take the lead in the development of an initial exploitation plan including a discussion on what each participant is hoping to gain from the project, and whether the expectations of all participants are compatible and coherent. This plan will be updated annually for the rest of the project. While the first exploitation plan is expected to be general and cover all activities in the project as a whole, the subsequent plans will focus more strictly of each of the identified exploitable results of the project.

The methodology to reach consensus on exploitation issues and to draw up a final exploitation plan is through **Exploitation Workshops**, described in detail in WP9. The exploitation strategy will answer crucial questions as:

- WHAT: clarification of exploitation results and development of exploitation routes.
- WHO: identification of lead partners to achieve each exploitable result and which project partners will exploit each exploitable result.
- HOW: which form(s) will the exploitation of these results take (direct industrial use, patenting, technology transfer, publications, etc.).
- BARRIERS: i.e. identification of (overlapping) IPR and commercialisation conditions - identification of risks actions to mitigate them; existing or emerging alternative solutions in competition with results; availability of knowledge and resources among participants and willingness to contribute to the result. IP issues and ownership of knowledge.

An exploitation plan will be prepared for each result and for the project as a whole and it will outline: i) Identification of all exploitable results & intellectual Property Rights (IPR) protection strategy.

- Analysis and evaluation of the exploitation potential of each result (market).
- Roadmap of exploitation activities during and after the project, exploitation routes.
- Technology implementation plan of the project results & the expected socio-economic impact.
- Factors influencing the exploitation (regulations, standards, etc.).
- Business plan for the key results. (cost-benefit feasibility, potential investors, transfer to other sectors)

Exploitation tracking: The use of Key Performance indicators will allow the consortium to monitor progress towards achieving the projects exploitation objectives for the target products. Table 2.2.e summarizes how the project will contribute to the KPI defined by the EC for Societal Challenges under H2020. More information about the contribution of each partner to the KPI (publications and conferences) see table 2.2.c.

Table 2.2.e DAFIA expected key Performance indicators (KPI) related to Societal Challenges.

DAFIA's partner	Patents and IPR protection	Products/technologies to develop.
AIMPLAS	1 Patent, Production of polyamides 56 from 100% renewable monomers. Trade secret with MINE to formulate PA56 with green flame retardants.	Polyamides 56 from renewable resources and fireproof bio-polyamides. New packages for fresh or frozen fish. Possibilities to use to pack other foods with similar requirements.
POLITO	1 Patent, on the use of the extracted/modified products as green/low impact flame retardants.	The recovery of biomacromolecules from wastes and their use in applications quite far from their standard use could represent a new product process/technology.
SFA	1 Patent shared with NUTRIMAR and SMC of gelatine extraction technology from untreated or pre-treated raw materials.	Several products (gelatine, gelatine hydrolysates) as well as extraction technologies covering modified/optimised gelatine extraction technologies from untreated or pre-pre-treated raw materials will be presented.
SMC	One with SFA and NUTRIMAR, on isolation of nucleic acids from fish RRM.	Two new processes, one on nucleic acid extraction from MRRM & one on 1,5-pentadiamine production from MSW.

DTU	1-2 Patent on engineering of yeast strains for production of *cis,cis*-muconic acid and adipic acid.	2 new technologies: microbial production of *cis,cis*muconic acid and adipic acid from complex feedstocks.
CNRS	1 or 2 Patents. One related to the sulfation of diacids in batch, the other related to the sulfation of di-acids in continuous flow.	Since the study will include a catalysts screening, several catalytic that could represent new products will be obtained
NUTRIMAR	1 patent application shared with SFA and SMC related to gelatine from fish-skin.	1-2 new products are expected from protein fraction and skin fraction of the by-products.
IRIS	1 Patent: combining fish gelatine with specific additives.	2 products will be launched post project: (1) barrier coating for packaging and (2) edible active coating for food, especially validated for fish application
BIOTREND	Trade secret or patent regarding downstream processing of adipic acid and/or *cis,cis*-muconic	Optimised strategies for the fermentative production and purification of adipic and/or *cis,cis*-muconic acid.

	acid from complex fermentation media.	
DLABS	2 patents on flame retardants based on protein and lignosulfonate macromolecules.	Two or three different FR macromolecules adapted for different plastics such as polyamides and polyesters.
MINE	Trade secret with AIMPLAS to produce new green polyamides containing nucleic base flame retardant. Direct sales or Licencing.	New range of bio-based polyamides and petrol based polyamides containing alternative flame retardant to APP.
BBEU	None envisioned	3 new processes: DSP development and scale-up of diamines; fermentation of diacids/diamines.
BIOPOLIS	1 Patent with the developed technology for pretreatment and fractioning of MSW. 1 patent on engineering microorganisms for the production of adipic acid.	Optimized strategies for the pre-treatment and fractioning of MSW, attending to minimizing energy costs and product spoilage. A new technology for the microbial production of adipic acid from complex feedstocks.
ARKEMA	1-2 patents. Mainly SH_2-muconic acid derivatives for polyamide modifications.	Thiol derivative. Diamine produced from muconic is a known product, but a new process.

Intellectual Property Rights protection activities are managed under WP9. The overall IPR strategy of the project is to ensure that partners are able to fully exploit their market position, involving patent protection.

The Partners will sign **a Consortium Agreement** (CA) prior to the signature of the Grant Agreement with the EC in which all the rules for ensuring confidentiality (procedure for granting publication of the results), ownership regime and IPR issues (background included and excluded, ownership, licensing terms, etc) are detailed. The CA will be based on the DESCA model. The CA will include the initial 'Key Exploitable Results' (Table 2.2.d) of the partners and patents interest list (Table 2.2.b), which will be the starting point for IPR protection and Exploitation Plans.

The following three areas of activity have been identified in order to ensure proper IP management and maximize the impact and market relevance of the knowledge generated in the project:

- Proper assessment of the pre-existing knowledge of the various project partners, their potential contribution to the project results IP, and potential overlap of IP. In shaping the IP strategy of the consortium these will be assessed based on the Consortium Agreement.
- The assessment report may be used to draft an IP and exploitation agreement, if needed, as an integral part of the final exploitation plan, in line with the consortium agreement, and used as an input for the exploitation plans. The document will be upgraded during the project to include new results unforeseen at the beginning of the developments as it is needed.
- The overall IPR strategy annually focuses on the knowledge generated by the project, the IP rights, the patentability and optimal IPR protection options. Any disputes over IPRs should be settled according to the consortium agreement. Legal requirements for commercialization are also explored, in particular through "freedom to operate" analysis as well as individual patent filing strategies and drafting of patent applications for the most promising project results, as it is needed.

As described in section 3.2, the SMEs partners will have a higher percentage of participation in the results exploitation, so the IPR management will respect at any time the objectives established in this type of Collaborative Project and thus watching over SMEs interests.

Exploitation through Standards development: Successful exploitation will only occur if the market is receptive to the products and processes developed in the project. Standards are a critical part of a functioning market place, standards for bio-based products are under development at CEN. After identification of the results with standardisation potential (initially focused on exploring the conversion routes of municipal solid waste (**MSW**), and marine rest raw-materials (**MRRM**) from the fish processing industries, to obtain **high added value products**), contacts will be made to relevant standardisation bodies in order to initiate, for instance a CEN workshop or contribute to existing standardisation activities in ISO. In the case of contributing to CEN, **DAFIA** will propose to CEN Workshop Agreement (CWA) a standardization document developed in a CEN Workshop (http://www.cen.eu/). All experts interested in developing the document are welcome in the development of such CWA. The CWA document will likely be a technical specification or a technical report, but could also become a broader guidelines document that includes information like guidance for continuous-reinforced plastic materials, best practice and information on material performances and production processes, etc. Several **DAFIA** partners have a strong link with standardisation bodies.

Post project execution of exploitation plan: As discussed a central aim of the DAFIA dissemination plan is **to 'prepare' the market in order to reduce the incubation time for DAFIA new products and technologies to a**

maximum of 5 years and to provide knowledge, methodologies and training materials that can contribute to increase competitiveness of the EU industry and workers skills in the field**,** facilitating the widespread adoption and market penetration of the developed materials and technologies.

Table 2.2.f summarises potential exploitation routes for the main key exploitable results. At the end of the project DAFIA technology will be at TRL5 and it is expected that a further 5 years will be required to achieve TRL8. Significant investment to move from TRL5 to TRL 8 is needed, i.e., additional budget outside of the project must be attracted using the patents and the trade secrets generated during the project and detailed in table 2.2.e.

Table 2.2.f Main target customers and knowledge licencing and sales strategy.

Product to be developed in DAFIA (and IPR owner)	***IPR licencing and sales strategy.***	***Main target Customers***
1. Cost-competitive technologies for the effective separation of components from complex waste (MSW &MRRM).		
Carbohydrates, proteins and lipids from MSW (BIOPOLIS).	Licensing, joint venture, commercial agreements with companies active in separation and downstream processes.	Companies interested in the valorisation of components of MSW or MRRM: Fish processing companies, MSW plants managers, local authorities, etc.
Proteins, nucleic acid and lipids from MRRM (SFA, SMC & NUTRIMAR).		
2. Production of intermediated products by microbial fermentation routes.		
Muconic/adipic acids (BIOTREND, DTU, & BBEU).	Licensing or production agreements with medium or large size fermentation companies (such as Verdezyne, Genomatica, Invista, DSM, Myriant, Bioamber, Amyris).	Current diacids and diamine users for polyamide, polyesters and diesters (solvents, plasticizers, lubricants).
1,5-Pentanediamine (SMC, BIOTREND & BBEU).		
3. Production of bioresins and additives by green chemistry routes.		
Gelatine based resins for active, barrier and/or edible coatings (IRIS AIMPLAS & NUTRIMAR).	Licensing agreements with coating production companies.	AIMPLAS, NUTRIMAR, as well as food and packaging converters, e.g. SALMAR, NESTLE, largest salmon producer in EU, was identified to have keen interest in both solutions.
Polyamides with adhesion promoters using SH_2-muconic acid derivatives (ARKEMA, ICERLYON).	Own ARKEMA production.	Plastic parts manufacturers of Automotive, electronics, industrial coating, consumer goods, ...

Bio-polyamide 56 (AIMPLAS & ARKEMA)	Licensing or production agreements with producers of polyamides (NUREL, UBE, Radici, etc)	Plastic parts and films manufacturer for packaging, automotive & Electric electronic sectors. Tier01 Automotive such as Faurecia, Johnsons control, Grupo Antolin, etc… Other manufacturer of polyamides parts: Schneider Electric, Simon, etc. Other sectors furniture, technical machinery, etc.
Nucleic acid based flame retardants (DLABS & POLITO).	Licensing or production agreements with chemical companies such as ICL, CLARIANT, BUDENHEIM.	ARKEMA, MINE and other compounding companies.
Bio-based polyamide compounds (ARKEMA, MINE & AIMPLAS).	MINE and ARKEMA will manufacture and direct sales to plastic products companies. Licencing to other compounding companies (out of the market scope of both companies).	Injection moulding companies that produce parts for automotive, furniture, building and electric electronic sector.

The partners have resources to cover the costs needed to implement the necessary co-funding for further developing and marketing the results of DAFIA. Besides internal resources in order to support the commercial exploitation of the project's results; EUREKA, national programs, investment funds (EU Investment Bank. EU Investment Fund), private investors and venture capital, SME instrument will be considered, as well as opportunities brought through Regional Funds (ESIF), which should be possible as DAFIA aligns perfectly to the priorities identified under the RIS3 strategy for most of the Partner's countries. To ensure that the maximum potential EU market is reached, the **IPREB** on behalf of the Consortium, will contact and involve Industrial Associations or SMEs in the new supply chain of DAFIA, all around EU thus achieving a multiplying effect and spreading the project results.

In summary, the protection of the interest of Industry, especially the SMEs, will be guaranteed as they are the

majority represented in the various boards, and consequently an industry-driven vote will be guaranteed in all levels of the management structure. The existing **consortium represents the complete research, development, manufacturing and material supply chain**, thereby **ensuring that the IPR protection procedure to be outlined within the Consortium Agreement** will be effectively managed and that the project results will be protected at each stage. DAFIA will result in a number of innovative processes, materials, products and IPR with a high potential for commercialization. Therefore this consortium considers the issues of IPR management and protection as particularly important and the PEDR will be prepared in order **to reduce the time to market** to the minimum possible; although further developments will be needed after the project end to achieve TRL8. There are no known business agreements or commitments which may impose limitations on the exploitation.

WP1: Characterisation, handling and pre-processing.	**WP Leader:** BIOPOLIS
Characterise and develop pre-processing methods ensuring availability and compatibility of the selected raw materials and their fractions for further fractionation and conversions in WP2 and WP3.	
WP2: Isolation and purification of products from pre-processed feedstocks.	**WP Leader:** SFA
Carry out a proof of concept to develop cost-efficient isolation and purification methods for nucleic acids and gelatine from the pre-processed feedstocks. The processes should enable utilisation of all residues, either as increased value-products or as fermentation substrates.	
WP3: Microbial conversions to diamines and diacids.	**WP Leader:** DTU
Develop robust microbial strains for production of chemical building blocks (diacids and diamines), by fermentation of pre-treated MSW and MRRM components. Fermentations will be optimized at laboratory scale. Downstream processing (DSP) for product extraction and purification will be developed. The most promising complete process (fermentation & DSP) will be tested at pilot scale in WP7 (BBEU).	
WP4: Development of Flame retardants from MRRM.	**WP Leader:** DLABS
Develop novel flame retardants based on nucleic acid and proteins extracted from MRRM in WP2 and on the chemical routes developed in BRIGIT project. The different flame retarding components will be tested separately and together in order to find optimal combinations to achieve the best balance between properties, cost and environmental impact. DAFIA aims to develop a multipurpose FR (similar to ammonium polyphosphate).	
WP5: Film and edible coatings from fish gelatine.	**WP Leader:** IRIS
The edible and barrier coatings formulations developed by IRIS in previous projects OliPHA and Wheylayer will be adjusted to gelatine from fish RRM, characterised and applied at pilot scale.	
WP6: Green chemical routes to produce polymers from developed building blocks.	**WP Leader:** AIMPLAS.

Production of polyamides from adipic acid and 1,5-pentanediamine, as well as the production of esters, nitriles, thiols and diols from muconic acid. The new chemical building blocks will be employed to improve the adhesion of polyamides on metal and/or glass.

The combined implementation of the proposed Communication and Dissemination Plan and all the IPR protection and Management procedures **will help to achieve the expected impact of the project** by facilitating the maximum market introduction of the project results whilst ensuring a significant impact on the general knowledge and skills of the related EU industry and scientific community through Open Access publications and contributions to standards, when possible. Consumers and general public will also be aware of the productivity and environmental benefits brought by **DAFIA** technologies, fostering the acceptation of research on the materials technologies, processing and final products validation as a worthwhile activity with direct impact on the EU socio-economic & environment status.

3.2. MANAGEMENT STRUCTURE AND PROCEDURES.

3.2.1. Organizational structure and the decision-making, including a list of milestones (Table 3.2a).

The Consortium of **DAFIA** will be coordinated by Partner AIMPLAS. The project management structure has been conceived to ensure the competent industrial, scientific and business management of **DAFIA** project and to enable efficient and effective integration and communication throughout the management chain. This management structure has been implemented successfully in similar EU Projects involving AIMPLAS, (e.g. FP7 BRIGIT, BUGWORKERS or BISIGODOS, and H2020 JOSPEL or URBANREC), and as such, has been adopted by the **DAFIA** Consortium. A schematic representation of the management structure of **DAFIA** is shown in Figure 3.2.a. Although the Coordinator is a Research Institute (AIMPLAS), the project management structure is clearly industry driven as the SME BIOTREND chairs the Executive Board (EB) and the SME Partner MINE chairs the IPR, Dissemination & Exploitation Board (IPREB). All boards have a majority representation of Industrial Partners as described below. An overview of the governance structure and decisionmaking process is also detailed below. The composition of the boards and the decision making mechanisms have been designed in order to ensure that all decisions are taken in accordance with the Industrial interests, and SME's interests in particular.

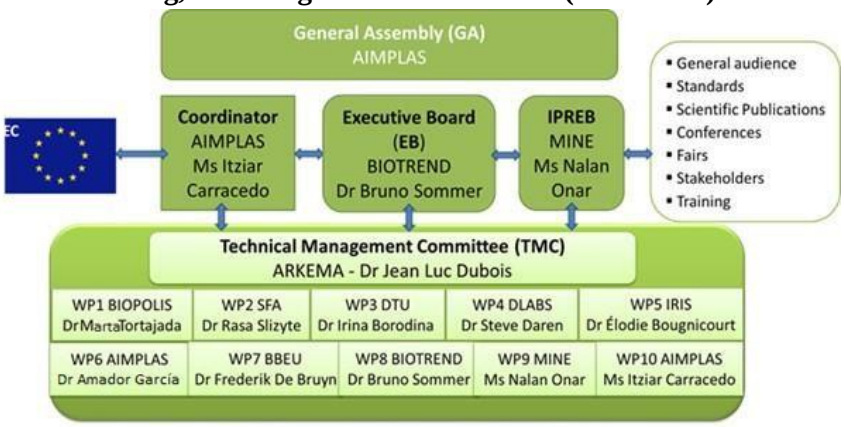


Fig. 3.2.a: Project Management Structure

The structure adopted is particularly business like (as opposed to bureaucratic) **to ensure that DAFIA is run as a "commercially" orientated project** with a very clear business focus. This has been achieved by limiting the number of partners represented on each management body, but ensuring that appropriate representation is pooled, in order to facilitate and speed up decision taking and ensure prompt reaction and implementation of countermeasures, if needed. Nevertheless, there is the opportunity for all partners to express their opinion and canvas support through the **General Assembly (GA)** which will meet at least twice a year to ratify progress and decisions made by the management bodies. The GA has the ultimate decision making power in the Consortium.

Decision –Making mechanisms: All Boards and Committees will typically meet at least twice per year. Any group will deliberate and decide validly provided a majority of its members are present or represented and each member will have one vote with the Chairperson having a casting vote if required. Decisions will be taken by a majority of two-thirds of the votes of the members present and the members that have delegated their vote. General Assembly meetings, meetings involving industrial demonstration or key technical meetings will be face-to-face, conference calls will be also encouraged.

All the partners have the responsibilities described below as a result of taking part in a specific decision making body**. In all of them, the SME and Industrial participants will have the highest level of decision-making competency (majority voting).**

The **General Assembly (GA)** consists of 16 representatives (one per partner), and is chaired by AIMPLAS. The GA is the ultimate decision-making and arbitration body of the Consortium (the final and highest authority in the project). However, it delegates short-term decisions-making actions to the **Executive Board** (EB). Such decisions do however require subsequent acceptance by the GA. The GA will decide on matters relating to: The political strategic and economic orientation of the Project, its objectives, definition and subsequent modifications, including the shift of budget of the Project. Actions towards defaulting parties, including their substitution. The nomination and election of EB members. The acceptance & exclusion of new parties or affiliates in the ECGA and in the Consortium Agreement. The Consortium Agreement and its Annexes. Description of Work, subsequent Project Implementation Plan and ECGA, including proposals and modifications. Ultimate approval of IPR Policies & Exploitation Plans proposed by the IPREB.

The **Executive Board (EB)** is responsible for the strategic and political guidance of the Project. It is **composed of one representative of each one of the 16 partners**. **BIOTREND** will chair the EB. The **responsibilities** of the **EB**

include: Overall guidance & supervision over strategic, political, economic & operational day-to-day issues regarding the Project. Definition, approval and fulfilment of Project objectives. Scientific and Technological Quality Assessment. Definition and implementation of risk management methodology at all levels (financial, technical and

administrative). Evaluation of results and definition of levels for dissemination relating to all kind of information and reports. Conflict Resolution. Constitution and administration of the funding. Presentation of the taken decisions to the GA periodically.

The EB will decide on matters relating to: The shift of tasks and responsibilities and budget of the Project. The regulations governing the decision-making and operational procedures of the EB and other Project bodies. The method of payment and management procedures of the Community's financial contribution. Requests for subcontracting, within the context of the Project. The actions to be taken in accordance with the evaluation of results obtained. The reporting, controlling and auditing procedures concerning the Project. The validation of reports and Project Deliverables to be submitted to the EC within the EC reporting activity together with annual validation of the realised expenditure in accordance with the budget of the Project.

The **Technical Management Committee** (**TMC**) is the body responsible for ensuring the overall technical coordination of the Project work, in accordance with the Work Programme, subsequent Implementation Plan, and EB decisions, at Scientific, Technical, Financial and Administrative level. The TMC's main responsibilities also include: ensuring co-ordination between WPs, facilitating the take-up of research results, supervision of the implementation of the EB decisions, and the assessment of Project results. The **TMC, Chaired by ARKEMA** will be composed by the WP Leaders: BIOPOLIS (WP1), SFA (WP2), DTU (WP3), DLABS (WP4), IRIS (WP5), AIMPLAS (WP6), BBEU (WP7), BIOTREND (WP8), MINE (WP9) and AIMPLAS (WP10).

The **TMC** will be responsible for: Ensuring the implementation of the decisions taken by the EB at WP level; Making proposals to the EB for the re-allocation of material, financial and human resources, tasks and responsibilities. Preparing deliverables & other information requested by the EB, based on reports by the WP Leaders. Validating and seeking EB approval for subcontracting. Ensuring co-ordination between WPs facilitating the take-up of research results in the demonstration activities. Assisting WP leaders on day to day management. Monitoring the projects technical progress against the identified technical risks, as described in T10.3.

The TMC will **convene** meetings with the Work Package Leaders at least **quarterly** (typically 3 face-to-face meetings and 2 conference call meetings) and if applicable with the EB, for purposes of co-ordination and management. Where the TMC cannot settle decision making disputes, the decision shall be taken in the EB.

The ***Work Package Leaders (WPL)***, as identified above will report and be accountable to the TMC. The specific WPL will be responsible for their own WP organizational arrangements, work procedures, day-to-day management and time schedule, providing that such arrangements, work procedures and time schedule do not conflict with the ECGA, the Consortium Agreement, the Description of Work, subsequent Implementation Plans, or requests and other provisions made by the appropriate bodies within the Consortium or the EC. Meetings of a specific WP will be convened at least quarterly, either face-to-face or by conference/video call. Extraordinary Meetings shall be: a) Called in and convened by the corresponding WP Leader; or b) by the TMC Chairperson.

The **Project Co-ordinator** (AIMPLAS) is the intermediate for communication between the Consortium and the EC and shall report and be accountable to the EB and the GA. The Co-ordinator is in charge of the financial and administrative management (WP10) of the Project. The Co-ordinator will submit for EB validation, the reports which shall be submitted to EC within the EC reporting activity. The reports covering project deliverables, supporting documents and audit certificates will be provided to the Co-ordinator by the TMC. For the avoidance of any doubt, the Co-ordinator has the capacity of representation of the consortium before the Commission. The Co-ordinator undertakes to keep the Joint Funds allocated by the EU Grant Agreement or by budget in accordance with the provisions which will be defined in the Consortium Agreement and in the Grant Agreement and decisions from the appropriate bodies. The main coordinators responsibilities will include:

a) Periodic Reporting: the project coordinator will be responsible for ensuring the collation and timely submission of the contractual reports required by the EC, always according to the EC reporting requirements for the H2020 Programme. With regards to the project periodic reports, the project manager's activities shall include: the creation of report /information gathering templates, providing support to the project consortium and enabling each project partner to undertake their responsibilities for the provision of information to complete such reports and making sure that each partner is made aware of the deadlines for the timely delivery of any required information. Project reports will be submitted to the EC on due time, including copies of the minutes, presentations given by partners and

additional information. **b) Periodic Commission Questionnaires:** if the Commission requires a number of periodic questionnaires, the Project Co-ordinator will be in charge of coordinating the completion and reporting of these questionnaires. **c) To ensure the verification and submission of the project deliverable reports:** all deliverable reports have to be verified initially by the EB before submission to the project coordinator, who will ensure that they are according to the EC requirements and submitted timely. **d) Coordination of the Periodic and Review Project Meetings:** the project coordinator will assume responsibility for the organization of those meetings, taking into account the attendance of the EC representative(s) and the independent reviewer(s) appointed by the EC. The project coordinator will coordinate: the meeting venue, agenda, minutes and other logistical arrangements where required,

in cooperation with the hosting Partner. **e) Collection of Certificates on Financial Statements:** the project coordinator will ensure that all Partners who are required to provide a CFS, provide them in a timely manner at the end of the Project. **f) Reporting and Payment of Project Financing:** assuming the role of project banker and taking responsibility for holding and payment of the project funds in accordance with the agreement of the General Assembly and the Consortium Agreement. **g) Additional responsibilities of the project coordinator will include:** coordination for the updating and signature of the project Consortium Agreement and Contract amendments (when required); to provide a platform for communication between the EC and the project consortium/ management structure; to provide a resource of support and information to the project consortium in fulfilment of the project contract and EC requirements.

The SMEs and Industrial Partners will own the ultimate decision making in accordance with the organizational structure. As detailed above, all Consortium bodies have a majority of Industrial Partners, thus, in the end, the decisions are taken by majority voting, guaranteeing that the participating Industries needs and requirements are fulfilled and consequently, that the results of the project will be for the benefit of the sectors involved in the **DAFIA** value chain.

From the Technical point of view, **the Project Co-ordinator, delegating day to day work to the defined Management Bodies, will monitor the overall progress of the Project in all its aspects,** with specific responsibilities in this regard for:

Development of the Technical Report Procedures and the Technical Progress Monitoring and templates for technical activities. Assuring interaction between WP's and technical quality of (intermediate) project results. Assuring WP interaction and also for the project scientific and technical quality assurance. Co-ordination, integration and validation of the Technical Reports and Project deliverables to be subsequently submitted to the EC. Co-ordination of the other committees' activities (IPR and Exploitation Board, EB, TMC).

The **IPR, Dissemination & Exploitation Board** (**IPREB**) will be chaired by partner MINE, and the rest of Members will be key persons from project partners who are involved in the innovation and IPR generation and protection of key technologies for their companies. The **IPREB** will have the responsibility to support the Consortium on IPR issues. It will propose the IPR Policy and the Action Plan for the Exploitation of project results to the EB and GA. The **IPREB** will be responsible for the preparation, monitoring and updating of the list of exploitable results and innovations that will be used to monitor the progress and achievements of the project and to assess the Plan for the Exploitation and Dissemination of Results (PEDR, including both the Dissemination and Business Plan). The **IPREB** will look to increase the IP protection and ownership awareness, assimilation and implementation, supported by training and workshops given by the IPR helpdesk and other EC support services that might be considered necessary. **In summary**, the protection of the interest of Industry, especially the SMEs will be guaranteed as they are the majority represented in the various boards, and consequently an industry-driven vote will be guaranteed in all levels of the management structure, allowing a *day-to-day management as in a processing chain at industrial level*, with protection of the SMEs interests and innovations.

3.2.2. Organizational structure and decision-making mechanisms.

The project management structure has been replicated from that used in FP7 projects coordinated by AIMPLAS. This structure has been proven to be agile and has been successful in properly managing projects of a similar nature to DAFIA, solving conflicts and overcoming hurdles for their successful completion. In addition, strong cooperation already exists among the majority of DAFIA Partners within FP6 and FP7 projects, with an open management and cooperative approach which benefits from the relevant experience of all Partners. Moreover, the project partnership is highly complementary and industry driven; each Partner with sound experience in their respective fields of work, have clear objectives and responsibility of tasks, and each of them fully understands and is highly committed to exploit the project outcomes, ensuring that the high degree of cooperation achieved already during proposal preparation will continue during project execution.

Innovation Management: the **IPREB** will implement the **Innovation Management strategy** developed within **DAFIA**. All DAFIA's technology can be exploited by a huge number of additional applications by modification of microbial strains, catalysts or green chemistry routes. The Partners complementarities along the value chain and their highly innovative nature will foster cross-fertilisation and boost innovation within the project, simply by the collaborative execution of the work plan. In addition, the project will constantly look for internal and external opportunities in order to further promote innovation. This way, as shown in figure 3.2.b, the project will use the dissemination and business plan in order to detect needs of potential customers and future end-users of the results and will proactively look for feedback from all stakeholders, entities and organisations (industrial platforms, public authorities, standard setters) that can be useful to improve **DAFIA** results, modifying accordingly the work plan if deemed useful. Similarly, the project partners will communicate internally the project development, fostering constructive criticism and feedback from staff, engineers, and designers and in particular managers. All this feedback

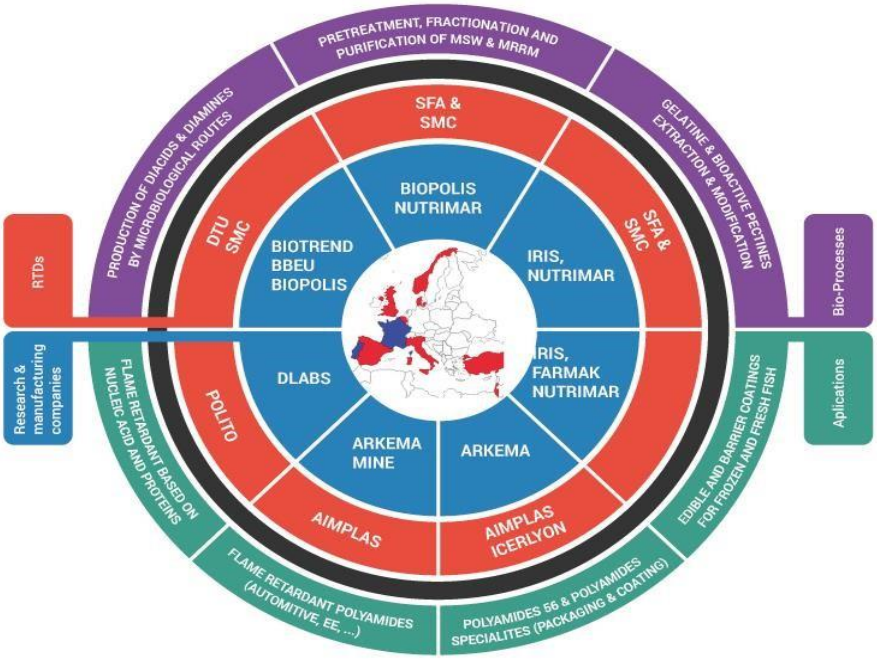
will be gathered by the **IPREB** and discussed during the **GA** meetings to agree possible modifications of the workplan if improvements in the results can occur resulting from this innovation management strategy.

3.3. CONSORTIUM AS A WHOLE

3.3.1. Overall description: adequacy, complementarities, representativity of the value chain and synergies.

DAFIA consortium has a well-balanced composition. The DAFIA consortium brings together a wealth of expertise and resources within the areas, including partners that propose the best solutions to maximize the valorisation of the different components of MRRM and MSW. The presence of main actors in the whole value chain demonstrates the critical mass of complimentary resources that will enable the DAFIA project to achieve its targeted societal, industrial, and scientific breakthroughs and commercial success. Each partner has a clearly defined role within the project and will contribute specific expertise that will enable the project success. **SME partners** (NUTRIMAR, IRIS, BIOTREND, DLABS, MINE, BBEU, BIOPOLIS and NNFCC) and the **IND partner** (ARKEMA) will provide the fundamental knowledge, technologies and expertise in the abovementioned areas required to achieve the project objectives, as their activities within the project are fully in line with their own core business and research strategies. Figure 3.2.a shows the role/ expertise of each partner in the project and their distribution around Europe.

RO Partners (POLITO, AIMPLAS, SFA, SMC, DTU and ICERLYON) have the necessary research expertise and will provide support to the partners to achieve their objectives. Each consortium member has demonstrated a long term commitment to the development of their own research, policy and commercial activities within the industrial targeted by the project.



All of them therefore offer excellent potential for a lasting integration of activities and the creation of a panEU culture for the technologies, materials and processes developed. The mobility of resources and personnel will be encouraged throughout the project, not only ***Figure 3.3.a.*** *Sectors and expertise brought by the DAFIA consortium* helping to form strong bonds

between the consortium members. Those actions are going to be implemented in order to lead towards an integration of complimentary activities, services and future research projects to improve the composite multi stage processing at EU level. This is a complete storyboard of actors complemented by an advisory board to extend industrial and technical insight.

The table below (Table 3.3.a) gives a brief overview of each partner specific business and their expertise, and the main tasks in the project allocated to them based on that expertise.

Table 3.3.a DAFIA's Participant Expertise and Tasks Distribution

Part num	Participant short name	Participant business / expertise-	Main role in the project
1	AIMPLAS	Research Technology Institute expert in polymer materials, nanotechnologies and processing.	Flame retardants polyamide compounds (lab-scale), production of polyamides 56, and coatings for packaging. LCA studies. Packages containing barrier coatings and bio-polyamides.

2	POLITO	Research University acknowledged as a center of excellence in Architecture and Engineering research areas.	Selection & characterization of biomacro-molecules classes (nucleic acid). Flame retardants characterization.
3	SFA	Technological expertise and industry knowledge in the utilization of renewable marine resources.	Fish and rest raw materials processing, extraction, fractionation, analysis and stabilisation of components. Gelatine extraction, protein hydrolysis
4	SMC	Institute with expertise in materials- and nanotechnology, oil and gas, chemical engineering, biotechnology and environmental sciences	Strain development for diamine production, small scale screening of strains and fermentation conditions in bioreactors up to 1-2L, nucleic acid extraction.
5	DTU	Research focusing on products that can be produced using bacteria, yeasts, and mammalian cells.	Strain development, uptake of carbohydrate & lipid fractions of complex raw materials to produce dicarboxylic acids and preliminary cultivation data.
6	CNRS	IRCELYON in the largest CNRS department devoted for fundamental and applied research in Catalysis and Environment in France.	SH_2-Muconic acids derivaties as adhesion promoters in polyamides.
7	NUTRIMAR	Bio marine company that produces oil, protein hydrolysate and meal from salmon rest raw materials.	Fish components fractionation. Barrier package for fresh fish slices.
8	IRIS	Advanced engineering company that specialises in process monitoring and control, and process optimisation.	Edible/ barrier coatings from fish gelatine. Active packages using active additives from MRRM.
9	BIOTREND	Research-based company focused on the development, optimization, de-risking and scale-up of bioprocesses for industrial biotechnology.	Process intensification through optimised feeding strategies, downstream process design for dicarboxylic acids and diamine, potential integration between fermentation and DSP.
10	DLABS	Research-based firm with capabilities in chemistry, polymers, biopolymers, synthesis and industrial processes.	Chemical modification of nucleic acid based flame retardant.
11	MINE	Production of petrol-based plastic compounds for several industrial sectors.	Flame retardant bio-based polyamides. Flame retardant compounds.
12	BBEU	Pilot plant for the development and scale up of new, bio-based and sustainable processes	Implementation of integrated dicarboxylic acid/diamine production process at pilot scale (1000L fermentation). Process protocol ready for demonstration scale operation.
13	BIOPOLIS	Biotechnology company offering contract research and manufacturing.	Municipal solid waste pre-treatment, extraction and validation for further fermentation activities. Strain selection and development for Adipic acid
14	ARKEMA	French chemical company that produce products from renewable and petrol based resources.	SH_2-Muconic acid chemical blocks to improve adhesion, polyamide flame retardant, polymerization of new polyamides.
15	NNFCC	Provide expert advice to business and the public sector on bioeconomy.	Communication, dissemination and business plan activities.

The participation and leadership of SMEs in DAFIA is clear as they represent the 50% of the Partnership and around 46% of the budget and 46% EC contribution**.** In section 3.2 (management structure and procedures), it can be seen that the relevant Project Governance Bodies (General Assembly, Executive Board, Technical Management Committee, Innovation Management Board and IPR-Exploitation Board) **have a majority of SMEs**, having the decision-making power in all of them. Also 6 of 10 WPs are led by SMEs.

To obtain the complementary range of expertise and the critical mass needed (both in human and financial terms), it is necessary to bring together, companies and ROs representing the entire supply chain and R&D areas, working in close cooperation to solve a common problem, which cannot be found within a single EU Region, requiring the integration of resources from across EU.

DAFIA partnership is composed by 15 organizations from 10EU countries, having the required pan EU approach essential for achieving the objectives of the project since it is required to address the EU social and Industry needs in the field with a wide EU perspective.

Furthermore, to ensure that the long-term and widespread take-up and commercial success of the DAFIA results, it is important that the project innovation and research activities are undertaken with a strong knowledge of the different industrial sectors addressed. The Consortium therefore ensures the participation of different EU regions, whose roles within the project are clearly defined and represent strategically important regional objectives, which will address the overall objectives of enhancing European industrial competitiveness, moving towards a circular economy and hence **providing a clear EU added value**.

3.3.2. Industrial/Commercial involvement to ensure exploitation of the results.

DAFIA **project clearly has a number of benefits for the involved SME** as we explain on detail along the proposal. It enables the transfer and market replication of the new materials for complex waste (MSW & MRRM) that contributes to the developed of four well-defined and high market products, supporting the take-up of the new advanced sorting, classification and production technologies and becoming a source of information, training and technical / regulatory support. Besides, DAFIA will create a new and more open market/supply chain for the materials, technologies and sales as previous figures shows.

The involvement of all the participants during and after the end of the project, in order to strengthen the industrial and commercial exploitation, is a key aspect of the consortium relationship, implemented through the innovation and IPR management strategies resulting in the Business Plan included in the Final PEDR.

During the project. Throughout the life of the project, all project partners will be responsible for updating the project 'Plan for the Exploitation and Dissemination of Results'. Dissemination will target 'take-up' actions within EU Local Authorities-Public Bodies, Stakeholders, SMEs and End-Users to promote the growth of the materials and technologies developed into new and existing market sectors; and to encourage SME participation within the DAFIA supply chain. Publication and dissemination of project knowledge and results will be conducted on agreement between the consortium partners, led by the **IPREB** chaired by MINE ensuring that the SMEs and industry interests are not only taken into account but are the real drivers for the exploitation strategy. Managed by the **IPREB**, each project partner will be required to develop their own 'Exploitation Plan' that will demonstrate how the project results will be integrated into their own forward business plan and research actions. Exploitation activities will include: protection and management of innovations, IPR and project results, gathering of reliable market intelligence and forecasts, feasibility studies for the creation of spin-off activities, evaluation of the project results for commercialization and application within new markets, steps to implement the proper industrial manufacturing policies, and the development of effective support structures to reinforce supply chain. Such activities are integral part of WP9.

After the end of the project. As explained on section 2.1, all the partners will obtain economic profits from the commercialization of the innovative DAFIA new materials and logistic implementation and will be able to introduce to the market cost competitive and sustainable products, increasing/opening markets and besides, to obtain profits by licensing the project results related to their knowledge field. However, as it will be detailed in the Business Plan and PEDR, the percentages of these profits will be different among the Partners depending on the exploitation agreement reached among co-owners which will reflect the percentage of contribution of each Partner during Project execution, the Background brought by them to achieve such result, and so on. Furthermore, these economic profits will encourage all the partners to contribute in the initial investment necessary to set up the exploitation process, once the project is finished. The SMEs participants are economically and managerially capable of performing the exploitation of the results, based on their turnover, their own investment in R&D and the level of economic contribution to the project. The other partners have also the right competences to guarantee the DAFIA exploitation of results in their areas. All partners are highly convinced of the potential impact of the project results in their businesses, and neither the SMEs, nor the other partners have conflict of interests regarding the areas of commercialization/exploitation, from a geographical, technological or market sector point of view. More details are provided in section 2.1., where apart from the exploitation interest, the steps to achieve the foreseen impact are described.

Industrial participation is very strong in this project. The consortium includes ten industry partners, with multiple and complementary technological, research and business expertise, covering the full spectrum of the activities proposed in DAFIA. They are dynamic enterprises with an active intention and outlook to expand their product/service portfolio into new markets. All DAFIA industrial partners planning to directly capitalize on the DAFIA results as we explain in detail in impact section, either by expanding their existing products/services, or by licencing technology. The geographical dispersion of the industrial partners ensures that they will cover an important proportion of European markets.

the DAFIA consortium there are three participants from the Associated Countries, Turkey and Israel, that as the Article 7 of the H2020 Regulation says are fully entitled to participate in this call for proposal with exactly the same right as they are considered eligible partners.

ANNEX II. Standard Composition of the MSW and MRRW and main expected conversion ratios of DAFIA products.

	MRRM (g/kg)	MSW (g/kg)	Diacids (g/kg MSW)	Diamine (g/kg MSW)	SH_2 derivatives (g/kg diacid)	C6 Diamine/ Dinitrile (g/kg diacid)	FR-PA 56 (g/Kg diacid/ diamine)	Flame retardant (g/kg Nucleic acid/protein)	Barrier Coating (g/kg gelatine)
Carbohydrates	---------	300							
Lipids	200-450	100	5	20	1000[a]	600[b]	850[c]		
Proteins	150-200	120						2,500[d]	
Nucleic acid	12-14								
Gelatine	10-40								1,500[e]
Bioactive Pectins	1-2								1-2 g/kg MRRM

[a] Since we add weight to the molecule (via SH_2), we can expect to keep a high mass yield even if the selectivity is not 100 %
[b] Muconic acid molecular weight is 142 g/mole. Adiponitrile molecular weight is 108 g/mole. Assuming 80 mol % yield, the mass yield is 600 g/kg of diacids.
[c] Assuming 85 mol % yield.
[d] The expected composition of the flame retardant is 20% nucleic acid, 20% Proteins, 40% Lignosulfonates, 20% other phosphorus based components.
[e] Gelatine will be modified with 50% of a fish or vegetal based plasticizer.

ANNEX III. Part A Figures and tables.

Task 5.2:

Expected role of each of the listed substances:

Type of action over the packed food	Type of additive to extract from MRRM	
	Polyphenols	Bioactive peptides
Antimicrobial/Antioxidant	X	X
Crosslinker	X	
Plasticizer		X

Task 6.1:

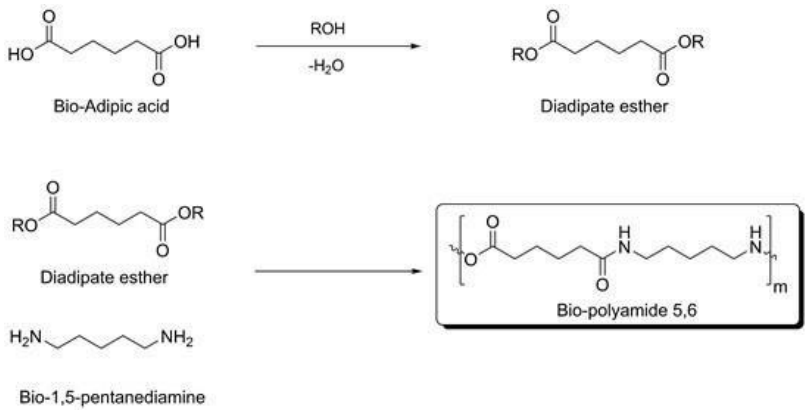


Figure 3.1.a Scheme of biopolyamide 5,6 synthesis-step 1 Task

7.3:

Case scenarios for the validation.

Type of Packaging	**Uncoated Fish**	**Fish with gelatine edible coating**	**Fish with gelatine edible active coating**
Unpacked		Frozen fillet	
Monolayer PET film		Frozen fillet	Fresh fillet
Monolayer PET film with gelatine barrier coating	Fresh fillet		Fresh fillet
Multilayer PET/EVOH/PE film with active coating	Fresh fillet		